Virtual University of Pakistan

BIO201 BIO202 BIO301 BIO302

MCQs for Midterms

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BIO201 MCQs Mids

Q

……energy is the energy of motion

(a) Kinetic

(b) Potential

(c) Chemical

(d) Atomic

Q

Ribosomes can be free or attached with

(a) golgi apparatus

(b) lysosomes

(c) endoplasmic reticulum

(d) necleus

Q

Which organ/molecule can convert light energy

(a) NAD

(b) Retina

(c) Biotin

(d) Coenzyme A

Q

Beta pleated sheet is found in

(a) Primary protein structure

(b) Secondary protein structure

(c) Tertiary protein structure

(d) Quaternary protein structure

Q

Inhibitor that binds to a site other than active site of enzyme

(a) Competitive inhibitor

(b) Irreversible inhibitor

(c) Reversible inhibitor

(d) Non competitive inhibitor

Q

lipid act as a following in which many components of cell membrane floats

(a) Pond

(b) Lake

(c) River

(d) Stream

Q

How many macromolecules a cell has....

(a) 2 (Maybe)

(b) 4

(c) 6

(d) 8

Q

Beta pleated sheet is found in

(a) Secondary protein structure

(b) Primary protein structure
(c) Tertiary protein structure
(d) Quaternary protein structure
Q
Cell membrane expels particles by vesicle formation
(a) pinocytosis
(b) Phagocytosis
(c) Endocytosis
(d) Exocytosis
Q
Molecule of phospholipids has ----------
(a) hydrophobic & Hydrophilic domain
(b) **hydrophobic only**
(c) hydrophilic only
(d) hydrogen linkage
Q
If one gene prevent the expression of other gene
(a) Pleiotropy
(b) Epistasis
(c) Synapse
(d) Tetrad
Q
Phenylalanine hydroxylase metabolizes
(a) Phenylalanine to Lysine
(b) Phenylalanine to Tyrosine
(c) Tyrosine to Phenylalanine
(d) Lysine to Phenylalanine
Q
Restriction enzymes inhibit the growth of
(a) Bacteriophage
(b) Bacteria
(c) Algae
(d) None
Q:
Missing of enzyme homogentisic acid oxidase caused
(a) Phenylketonuria
(b) Parkinson disease
(**c) Alkaptonuria**
(d) Cystic fibrosis
Q
Microtubules are originated from
(a) Cytoskeletal
(b) Nucleus
(c) Proteins
(d) Centrosomes
Q

The disintegration and condensation of nuclear material takes place in
(a) prophase II
(b) anaphase II
(c) metaphase II
(d) telophase II
Q
RNAs in large subunit of Eukaryotic ribosome
(a) 1
(b) 2
(c) 3
(d) 4
Q
Centromeres separate during whose anaphase:
(a) Mitosis
(b) meiosis I
(c) meiosis II
(d) None
Q
Enormous genetic information is present in
(a) RNA
(b) DNA.
(c) Nucleolus
(d) Nucleoplasm
Q
Okazaki fragments are joined by
(a) Ligase
(b) Gyrase
(c) Helicase
(d) Primase
Q
How much water present in a cell.....
(a) 50%
(b) 60%
(c) 70%
(d) 80%
Q
Charge on DNA is
(a) Positive
(b) Negative
(c) Neutral
(d) None

Q
Which of the following stabilizes cell shape
(a) DNA
(b) RNA

(c) **Actin**
(d) Myosin
Q
……energy is the energy of motion
(a) Kinetic
(b) Potential
(c) Chemical
(d) Atomic
Q
Space between double membrane structure of endoplasmic reticulum is called
(a) Vesicles
(b) Leumen
(c) Cavity
(d) **endoplasmic space**
Q
----------is one of the example of co-factor
(a) Copper
(b) Biotin
(c) Haeme
(d) All
Q
lipid act as a following in which many components of cell membrane floats
(a) Pond
(**b) Lake**
(c) River
(d) Stream
Q
DNA replicates during.....
(a) G0 phase.
(b) G1 phase
(**c) G2 phase**
(d) S phase
Q
Difference between the alpha 1-4 glycosidic and beta 1-4 glycosidic bonds is on the basis of :
(a) Specificity
(b) Stability
(c) neutral pH
(d) size and shape
Q
Example of variable expressivity
(a) Dementia
(b) Huntington disease
(c) Alzheimer disease.
(d) Pancreatic necrosis
Q
Following has a major role in down syndrome

(a) Maternal age
(b) Fraternal age
(c) Child age
(d) Relative age
Q
DNA replication obeys
(a) Dispersive model
(**b) Conservative model**
(c) Semi-conservative model
(d) Distractive model

Q
Missing of enzyme homogentisic acid oxidase caused
(a) Phenylketonuria
(b) Parkinson disease
(c) Alkaptonuria
(d) Cystic fibrosis
Q
The RNA is --------- while DNA is ----------
(a) double stranded, double stranded
(b) single stranded, double stranded
(c) double stranded, single stranded
(d) partially double stranded, double stranded
Q
Ribosome after protein formation arrives at
(a) Lysosome
(b) Rough endoplasmic reticulum
(c) Smooth endoplasmic reticulum
(d) Golgi bodies
Q
The nucleus membrane is continuous with
(a) golgi apparatus
(b) ribosomes
(c) endoplasmic reticulum
(d) nucleus
Q
Power generating organelle of cell is:
(a) golgi bodies
(b) nucleus
(c) mitochondria
(d) cell membrane
Q
The disintegration and condensation of nuclear material takes place in
(a) prophase II
(b) anaphase II
(c) metaphase II

(d) telophase II
Q
An observable feature in the subject of genetics is called
(a) Trait
(b) Character
(c) Phenotype
(d) All
Q
Ribosome after protein formation arrives at
(a) Lysosome
(**b) Rough endoplasmic reticulum**

202 MCQs Mids BIO

Q
Glucose residues in amylose are linked by _____.
α (1,4) linkage
β (1,4) linkage
α (1,6) linkage
None of these

Q
The number of carbon atoms in lysine is _____.
4
6
8
10

Q
If the carbonyl group is present at the end of the monosaccharide then it is called _____.
Acid anhydride
Aldose
Ketose
None of these

Q
The number of stereoisomers for a molecule containing only one chiral carbon is _____.
16
2
4
8

Q
Seminal fluid is rich in _____.
Fructose
Lactose

Xylose
Arabinose

Q
Which of the following glycosidic linkage exists between two glucose units of maltose?

alpha 1,4
beta 1,4
alpha 1,6
alpha1,4 and alpha 1,6

Q
Amino sugar is formed by the replacement of the hydroxyl group at ------ of the parent monosaccharide with amino group.
C-1
C-2
C-3
C-6

Q
Which one of the following is a structural homopolysaccharide?
Cellulose
Starch
Glycogen
All of these
Q
Which one of the following amino acids is not specified by three letter codon?
Serine
Proline
Selenocysteine
Methionine

Q:
Acid dissociation constants are designated as _____.

PKa
K
Pa
None of these

Q
The pI of glutamate is _____.

1.2
5.6
3.22

6.7

Q
Which of the following amino acid is negatively charged at physiological pH?
Aspartate
Lysine
Alanine
Proline

Q
Which of the following refers to particularly stable arrangements of amino acid residues giving rise to recurring structural patterns?
Primary structure
Secondary structure
Tertiary structure
Quaternary structure

Q
Keratin is present in _____.
Hair
Nail
Hoof
All of these

Q
Which of the following amino acid is involved in the hydrophobic interactions in the tertiary structure of protein?
Phenylalanine
Cysteine
Glycine
None of these

Q
The free energy of a protein molecule is influenced by _____.
Hydrophobicity
hydrogen bonds
electrostatic interactions
All of these

Q
Which of the following diseases is caused by protein misfolding?
Alzheimer's disease
Huntington's disease
Parkinson's disease
All of these

Q
Iron is a _____.

non-metal
metalloid
transition metal
halogen

Q
Myoglobin can bind _____ molecules of oxygen.

One
Two
Three
Four
Q
The partial pressure of oxygen (pO2) in lungs is about _____.
30 mmHg
50 mmHg
100 mmHg
200 mmHg

Q
In sickle cell anemia, Glutamate is replaced with _____.

Arginine
Alanine
Methionine
Valine

Q
Oxy hemoglobin dissociation curve describe the relationship of ----------
Available Oxygen
Amount of carbon dioxide carried by hemoglobin
Fibrinogen
Oxy-hemoglobin curve stability
Q
Which one of the following is made up of two molecules of glucose?
Maltose
Starch
Cellulose
Amylose

Q
Which bond exists between the carbonyl group and oxygen of a hydroxyl group in ring structure of glucose?

Ionic
Covalent
hydrophobic interaction
both ionic and covalent

Q
In Benedict test, the formation of cuprous oxide is indicated by the formation of _____.
blue ppt
silver mirror
red ppt
none of these

Q
Glucose 6 Phosphate is formed by the addition of phosphate group to C6 of glucose by _____ linkage.
Ether
Ester
Ionic
none of these

Q
E. coli convert lactose of milk to lactic acid by action of _____.
Beta galactosidase
Amyloglucosidase
Alpha amylase
Maltase

Q
At any pH above pI, gycine has a net _____ charge and will move towards the _____.
negative, anode
negative, cathode
positive, anode
positive, cathode

Q
Which of the following amino acids contain two COOH groups?
Alanine
Glutamate
Glycine
Valine

Q
Which of the following amino acid is positively charged at physiological pH?
Lysine
Aspartate
Tyrosine

Alanine

Q
Peptide bond is _____ linkage between two amino acids.
Ester
Amide
Ether
None of these

Q
Which of the following shows the greatest tendency to form α helices in most experimental model systems?
Glycine
Alanine
Proline
Serine

Q
Immunoglobulin G (IgG) consists of _____.
Two heavy chains and two light chains
Four heavy chains
Four light chains
None of these

Q
The free energy of a protein molecule is influenced by _____.
Hydrophobicity
hydrogen bonds
electrostatic interactions
All of these

Q
Proteins are denatured by _____.
Temperature
Ph
organic solvents
temperature, pH, organic solvents

Q
About _____ of amino acid residues in the myoglobin are found in alpha helices.
98%
78%
27%
10%

Q

Hemoglobin is found exclusively in _____.
white blood cells
red blood cells
platelets
platelets and white blood cells

Q
The partial pressure of oxygen (pO2) in lungs is about _____.
30 mmHg
50 mmHg
100 mmHg
200 mmHg

Q
In sickle cell anemia, Glutamate is replaced with _____.
Alanine
Methionine
Valine
Arginine

BIO301 MCQs Mid

Q
Genetics can be studied by
One way
Two ways
Three ways
Four ways

Q
Example of Mendilian genetics
Gene linkage
Pliotropy
Epigenetic
None

Q
In nature variation is
Temporary
Consistent
Widespread
All

Q
Condition in which an intermediate phenotype is developed from cross of parental generation.
Co dominance
Incomplete dominance
Both
Over dominance

Q
Condition in which one gene prevent the expression of other gene
Polymorphism
Pleiotropy
Epistasis
Polygon

Q
Bacterial chromosome is made up of
DNA
Histone
Both
RNA
Q
structure for chromosome stability
Telomere
Centrosome
Centromere
Chiasmata

Q
Broken ends of chromosomes are
Fluffy
Sticky
Transparent
Non sticky

Q
Centromere is unable to work due to mutation in
CDE-I
CDE-II
CDE-III
CDE-IV

Q
XO system of sex is present in
Fuitfly
Grasshoper

Snake
Frog

Q
prokaryotic cell divides by
Budding
Grafting
Mitosis
Binary fission

Q
Chromosomes become fit in the central line
Anaphase
Metaphase
Telophase
Prophase
Q
Haploid cells are produced by
Mitosis
Meiosis
Both
Parthenogenesis

Q
Tetrads align at center
Telophase I
Prophase I
Metaphase I
Anaphase I
Q
Which one is true
Sister chromatids are not identical in mitosis
Sister chromatids are identical in meiosis
Sister chromatids are not identical in meiosis
Non sister chromatids are identical in mitosis

Q
Genes are present on chromosomes first time said by
Greogor Mendel
T.H.Morgan
Walther Flemming
T.J.Robert

Q
Cross for determination of genotype of an individual

Cross over
Hybrid cross
Dihybrid cross
Test cross

Q
In ABO blood group system
Allele A is dominant over allele B
Allele B is dominant over allele A
Alleles A and B show co-dominance
Alleles A and B show incomplete dominance

Q
If two genes are more close then frequency of cross over will be
High
Low
High and low
Medium

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Q:
The hydrophilic backbones of alternating deoxyribose and phosphate groups are inside the double helix, facing the surrounding water?
a) True
b) False
c) False in certain cases
d) a and c
Q:
Chromosomes are made up of.....
a) Protein and DNA
b) DNA and RNA
c) RNA and protein
d) a & c
Q:
Chromosome theory of inheritance by Thomas Hunt Morgan was presented in
a) 1913
b) 1912
c) 1911
d) 1910
Q:
Nucleotide consists of …… major molecules
a) 2
b) 3
c) 4
d) 5

Q:
How many amino acids are there in one compelete tern of DNA...
a) 6.3
b) 3.6
c) 36
d) 4.6
Q:
Nucleosome fold to form the _____ fiber
a) 20 nm
b) 30 nm
c) 40 nm
d) 50 nm
Q:
On which media, cells are preferred to be grown?
a) N14
b) Pb
c) Cs
d) Ra
Q:
DNA polymerase has _____ domains
a) 3
b) 4
c) 5
d) 6
Q:
An E.Coli chromosome is estimated to have about.....super-coiled loops
a) 500
b) 400
c) 300
d) 200
Q:
Separation of DNA is accomplished by _____.
a) Type I topoisomerase
b) Type II topoisomerase
c) Type III topoisomerase
d) DNA polymease
Q:
Which one of the following is not an ionizing radiation?
a) X-rays
b) Ultraviolet light
c) Gamma rays
d) β rays
Q:
Hydrolytic deamination of cytosine is estimated to take place about
a) 1000 to 5000 times a day in a mammalian cell
b) 100 to 500 times a day in a mammalian cell

c) 10 to 50 times a day in a mammalian cell
d) 1to 5times a day in a mammalian cell
Q:
What will happen to the DNA exposed to N-methylguanine
a) it will be oxidized
b) it will make dimers of the bases
c) it will lead to production of mono-adduct
d) it will form a basic site
Q:
Aflatoxins are produced by
e) Bacteria
f) Virus
g) Fungi
h) Parasites
Q:
.....are similar enough to the proper bases to get taken up by cells converted into nucleoside triphosphate and incorporated into DNA during replication.
a) Interclating agents
b) Base analogs
c) Amino acid analogs
d) None of these
Q:
Cells infected with phage .....provide the ideal system to find the true template for protein synthesis
a) T4
b) P4
c) S4
d) L4
Q:
RNA polymerase performs essentially the same reaction in the cells from bacteria to human that is the synthesis of .....
a) cDNA
b) DNA
c) RNA
d) DNA, RNA and Cdna
e) Q:
There are three human MutS homologs
a) MSH1, MSH2, MSH3
b) MSH2, MSH3, MSH4
c) MSH2, MSH3, MSH6
d) MSH2, MSH3, MSH5
Q:
In long patch repair, how many nucleotides are repaired...
a) 1
b) 4-6

c) 2-8
d) 10-14
Q:
Pyrimidine dimers are formed by...
UV radiation
Infra red
Ultrasound
nuclear radiation

http://https://chat.whatsapp.com/FmGdNtjc2pMEpCHnzpVjdO

Q

TH2 cells mainly help to activate :

a) T cells
b) B cells
c) MHC I molecules
d) MHC II molecules

Q
Which of the following stabilizes cell shape
a) DNA
b) Actin
c) Myosin
d) RNA

Q

Fibroblast cells can be induced to proliferate by treatment with .....
a) Platelet-derived growth factor (PDGF)
b) Epidermal growth factor (EGF)
c) Both (EGF) and (PDGF)
d) None

Q
Even in the absence of antigen humans can make more than..... different antibody molecules.

a) 10^{120}
b) 12^{10}
c) 10^{12}
d) 10^{112}

Q

The _____immune system, like the ____system, can remember prior experiences

a) Adaptive, nervous

b) Adaptive, respiratory

c) Innate, nervous

d) Activated, metabolic

Q

Inhibitor that binds to a site other than active site of enzyme

a) Competitive inhibitor
b) Reversible inhibitor
c) IRReversible inhibitor
d) Non competitive inhibitor

Q

Condition in which cells have specific binding

a) cell recognition
b) cell communication
c) cell adhesion
d) a & c

Q

In circular form, glucose molecule can adapt alpha or beta configuration. In beta glucose, OH group is pointing ------ while in case of alpha OH group is in --------- direction

a) Upward, downward
b) Downward, upward
c) Upward, linear
d) Upward, linear

Q

People with additional Y chromosome have external appearance

a) Abnormal
b) Normal
c) None
d) May be normal or abnormal

Q

Which one is the product of genetic engineering

a) Insulin
b) Quina
c) Plasmid
d) Fatty acids

Q

Which one is initiation codon

a) UGA
b) AUG
c) UAG
d) UAA

Q
Function of Golgi apparatus is

a) Packaging and transportation of proteins
b) egradation of proteins
c) Homeostasis
d) Cell movement

Q
An average globular cell have ------ mitochondria

a) 100,000
b) 10,000
c) 1,000
d) 100

Q
Chromosome fibres consist of

a) DNA and protein
b) phosphate bonds
c) nucleosomes
d) protein only

Q
Microtubules which are not attached to the chromosomes are

a) Mitotic Centre
b) Centrosomes
c) Polar microtubules
d) Polar microtubules

Q
The number of appearance of chromosomes in the nucleus of a eukaryotic cell is called

a) Prototype
b) Genetic makeup
c) Karyotype
d) Karyotype

Q
HIV is

a) DNA virus
b) RNA virus
c) Herpes virus
d) Adeno virus

Q

Structure formed by many folding of polypeptide chain

a) Primary protein structure
b) Secondary protein structure
c) Tertiary protein structure
d) Quaternary protein structure

Q
Ribosomes are engaged in:

a) Lipid synthesis
b) Protein synthesis
c) DNA synthesis
d) Sugar synthesis

Q

_____ are removed during RNA maturation.

a) Introns
b) Exons

c) Intron-Exon boundaries

d) Promoter region only

Q
TH2 cells mainly help to activate :

a) T cells
b) MHC I molecules
c) B cells
d) MHC II molecules

Q
Mendel donated dominant and recessive traits as
a) Ss and ss
b) SS and ss
c) ss and Ss
d) SS and SS

Q
Structure of protein is
a) One dimensional
b) Two dimensional
c) Three dimensional
d) Four dimensional

Q
The fusion of membranes from same compartment is termed as
a) Heterotypic fusion
b) Homotypic fusion
c) Hemitypic fusion
d) Hemitypic fusion

Q
Which one of the following substances in immune system causes capillaries to dilate and become leaky?

a) Amino acids
b) Thymine
c) Carbohydrates
d) Histamine

Q
On which site a substrate binds on an enzyme.....
a) Peptidal site
b) Active site

c) Binding site
d) Catalytic site

Q
For optimal activity of hydrolytic enzymes, lysosome maintains a pH about
a) 7.0
b) 5.0
c) 3.0
d) 9.0

Q
Which one of the following helps in stimulating breast cells to produce milk?
a) STAT 1
b) Erythropoietin
c) Prolactin
d) Interferon

Q
Proteins which are fully translocated across Endoplasmic reticulum and are embedded in it
a) Transmembrane proteins
b) Water soluble proteins
c) Extrinsic proteins
d) Extrinsic proteins

Q
Activation energy of an enzymatic reaction is
a) Low
b) High
c) Intermediate
d) All

Q
Ran GTPase is a molecular switch that exist in
a) 2 states
b) 6 states
c) 8 states
d) 4 states

Q
----------is one of the example of Coenzyme
a) Iron
b) Zinc
c) NAD
d) Haeme

Q
An oncogene after mutation produces one of abnormal condition

a) Kidney stones
b) Encephalopathy
c) Diabetes
d) Cancer

Q
Enzyme used in PCR process

a) *Thermus hydrophila*
b) *Thermus punctata*
c) *Thermus aquaticus*
d) *Thermus fario*

Q
Bacterial DNA is present in
a) Nucleoid
b) Nucleolus
c) Nucleus
d) Nucleoplasm

Q
Dendritic cells use on of following signal to guide helper T cells to effector T cells
a) IL-4
b) IL-3
c) IL-2
d) IL-1

Q
Genes are located in the nucleus and tangled charamatin material wrapped around histones in .....
a) Both Eukarotes and Prokaryotes
b) Eukaryotes
c) Prokaryotes
d) None of these

Q
Organisms shuffle genes by
a) Meiosis
b) Mitosis
c) binary fission
d) multiple fission

Q
Inactive "Intracellular Receptors" (ICR) may occur .....

a) free moving in plasma
b) Occur in cytoplasm
c) Occur as DNA bound
d) Both (Occur in cytoplasm) and (DNA bound)

Q
The cells which are activated initially by recognizing foreign antigen on APC
a) B cells
b) Macrophages
c) B and T cells
d) T cells

Q
a) Lipids are……..
a).... insoluble, hydrophobic molecule
b) soluble, hydrophilic molecules
c) basic molecule
d) acidic molecule

Q
How many times an average cell divide before it gets unhealthy
a) 25 times
b) 50 times
c) 75 times
d) 100 times

Q

_____ are removed during RNA maturation.
a) Introns
b) Promoter region only
c) Intron-Exon boundaries
d) Exons

Q
T.H.Morgan observed

a) Independent assortment of genes
b) Gene linkage
c) Segregation of genes
d) Fertilization

Q
Cholera toxin prevents.....

a) Hydrolysis of GTP to GDP
b) Formation of ATP

c) Formation of GTP
d) None of these

Q
Which one is larger sex chromosome
a) X
b) Y
c) XY
d) None

Q
Stitching two different DNAs together is
a) Replication
b) Genetic engineering
c) Translation
d) Transcription

Q
Philadelphia chromosome is a chromosomal translocation between chromosome ___ and _____.
a) 5 & 6
b) 7 & 21
c) 1 & 2
d) 9 & 22

Q
Cyanobacteria are type of bacteria that have
a) Cyanocide
b) Nucleus
c) Chlorophyll
d) Mitochondria

Q
The _____immune system, like the ____system, can remember prior experiences
a) Adaptive, respiratory
b) Adaptive, nervous
c) Innate, nervous
d) Activated, metabolic

Q
Chromosome fibres consist of
a) protein only
b) nucleosomes
c) DNA and protein
d) phosphate bond

Q
How many types of bacterial colony were used by Freddrick Griffith in his experiment

a) 2
b) 4
c) 6
d) 8

Q
Endoplasmic reticulum is composed of rough and smooth endoplasmic reticulum and plays important role in ----------

a) Sorting and modifying protein
b) Sorting protein
c) Modifying protein
d) Degrading protein

Q
There are ..... lymphocytes in the human body, making the immune system comparable in cell mass to the liver or brain

a) 2×12^{10}
b) 2×10^{21}
c) 2×10^{12}
d) 12×10^{12}

Q
Genes are located in the nucleus and tangled charamatin material wrapped around histones in .....

a) Both Eukarotes and Prokaryotes
b) Eukaryotes
c) Prokaryotes
d) None of these

Q
If one gene prevent the expression of other gene

a) Pleiotropy
b) Epistasis
c) Synapse
d) Tetrad

Q
The fusion of membranes from same compartment is termed as

a) Heterotypic fusion

b) Homotypic fusion
c) Hemitypic fusion
d) Hypotypic fusion

Q
Approximately, How many years ago life orignated.....
a) 4,000,000,000
b) 4,000,000,000, 000,000
c) 4,000,000,000,000
40,000,000,000

Q
Proteins which are fully translocated across Endoplasmic reticulum and are embedded in it
a) Intinsic proteins
b) Extrinsic proteins
c) Water soluble proteins
d) Water soluble proteins

Q
Phenylalanine hydroxylase metabolizes
a) Phenylalanine to Lysine
b) Phenylalanine to Tyrosine
c) Tyrosine to Phenylalanine
d) Lysine to Phenylalanine

Q
Activation of Phospholipase C-β
a) 10%
b) 90%
c) 20%
d) 1%

Q
Mental retardation is caused by
a) Leishmaniasis
b) Cystic fibrosis
c) Phenylketonuria
d) Sickle cell anemia

Q
The agent that is used as cryoprotectant is

a) KMNO4
b) Acetic acid
c) EDTA
d) DMSO

Q:
cell membrane intake particles by vesicle formation
a) Endocytosis
b) Exocytosis
c) Pinocytosis
d) Phagocytosis

Q
In 2014 it was reported that one of following proteins in blood is helpful in cure of Alzheimer's disease and heart disease
a) A 11
b) A12
c) A13
d) A14

Q
Undefferentiated cells of multicellular organisms which can develope into specialized cells
a) Hepatocytes
b) Degenerative cells
c) Mast cells
d) Stem cells

Q
Autocrine signals bind to the receptors on the cells, which .....
a) are arround them
b) has no involvement of the receptors
c) Adjacent Cells
d) Cells that secrete them

Q
People with additional Y chromosome have external appearance
a) Normal
b) Abnormal
c) May be normal or abnormal
d) None

Q
Helices that are/is involved in agonist binding are
a) Helix 6
b) Helix 5
c) Helix 3
d) All

Q
The lipids molecules held together by weak interactions and
a) Van der waals forces
b) Hydrogen bonds
c) Isotonic bonds
d) Esterase bonds

Q
β-blockers (antagonist) are involved in
a) Bind to receptor by not activating it
b) None of these
c) Mimicking the function of a hormone
d) Making the hormone activity better

Q
a) Initiation site of transcription
b) Co factor
c) Promoter
d) Active site
e) Effective site

Q
Sex related disorders are
a) X- related
b) Y- related
c) XY- related
d) XYZ - related

Q
Which one is initiation codon
a) UAG
b) AUG
c) UGA
d) UAA

Q
Function of G protein in vision may involve.....

a) Removal of NADPH
b) Production of ATP
c) Inactivation of K+ channels
d) Activation of Phospholipase C-β

Q
Calmodulin is about.....% of total cell protein mass
a) 10%
b) 20%
c) 90%
d) 1%

Q
The agent that is used as cryoprotectant is
a) Acetic acid
b) KMNO4
c) DMSO
d) EDTA

Q
In prometaphase I, only ONE of the following occurs:
a) Sister chromatids start pairing up
b) Centrosomes start moving to the opposite poles and form microtubules
c) Condensation of DNA material
d) Disintegration of the nuclear membrane

Q
In 2014 it was reported that one of following proteins in blood is helpful in cure of Alzheimer's disease and heart disease
a) A11
b) A12
c) A 13
d) A 14

Q
Undefferentiated cells of multicellular organisms which can develope into specialized cells
a) Mast cells
b) Hepatocytes
c) Stem cells
d) Degenerative cells

Q
Once the genetic information has passed on to the next generation, it does not
a) change

b) inherit to next generation
c) remove
d) remove

Q
Both (Occur in cytoplasm) and (DNA bound)
a) Addition of phosphate group
b) Removal of phosphate group
c) Both addition and removal and thrionine residues of phosphate group
d) Phosphorylation of specific serine

Q
The fusion of membranes from same compartment is termed as
a) Heterotypic fusion
b) Hemitypic fusion
c) Homotypic fusion
d) Hypotypic fusion

Q
For optimal activity of hydrolytic enzymes, lysosome maintains a pH about
a) 3.0
b) 3.0
c) 5.0
d) 7.0

Q
Phosphorylation signaling in the E. coli alters
a) Kinase acctivity
b) Transcription factor
c) Responder function
d) Protein synthesis

Q
Proteins which are partly translocated across Endoplasmic reticulum and are embedded in it
a) Transmembrane proteins
b) Water soluble proteins
c) Intrinsic proteins
d) Extrinsic proteins

Q
Ran GTPase is a molecular switch that exist in
a) 2 states
b) 4 states
c) 6 states
d) 8 states

Q
How many protein molecules an animal cell contains
a) 5 billion
b) 10 billion
c) 15 billion
d) 20 billion

Q
Autocrine signals bind to the receptors on the cells, which .....
a) are arround them
b) Cells that secrete them
c) Adjacent Cells
d) has no involvement of the receptors

Q
An oncogene after mutation produces one of abnormal condition
a) Diabetes
b) Kidney stones
c) Encephalopathy
d) Cancer

Q Dendritic
cells use on of following signal to guide helper T cells to effector T cells
a) IL-1
b) IL-3
c) IL-4
d) IL-2

Q
The cells which are activated initially by recognizing foreign antigen on APC
a) B cells
b) B and T cells
c) Macrophages
d) T cells

Q
_____are passed on to the daughter cells after mitosis but not passed on to the children.
a) Germ line mutations
b) Point mutations
c) Somatic mutations
d) Chromosomal mutations

Q
Philadelphia chromosome is a chromosomal translocation between chromosome ___ and _____.
a) 7 & 21
b) 1 & 2
c) 9 & 22

d) 5 & 6

Q
The _____immune system, like the ____system, can remember prior experiences

a) Adaptive, nervous
b) Adaptive, respiratory
c) Innate, nervous
d) Activated, metabolic

Q
Even in the absence of antigen humans can make more than..... different antibody molecules.

a) 10^{120}
b) 12^{10}
c) 10^{12}
d) 10^{112}

Q
Undefferentiated cells of multicellular organisms which can develope into specialized cells

a) Mast cells
b) Stem cells
c) Hepatocytes
d) Degenerative cells

Q
For optimal activity of hydrolytic enzymes, lysosome maintains a pH about

a) 3.0
b) 5.0
c) 7.0
d) 9.0

Q
Ran GTPase is a molecular switch that exist in

a) 2 states
b) 4 states
c) 6 states
d) 8 states

Q
How many protein molecules an animal cell contains

a) 5 billion
b) 10 billion
c) 15 billion
d) 20 billion

Q
There are ..... lymphocytes in the human body, making the immune system comparable in cell mass to the liver or brain

a) 2×12^{10}
b) 2×10^{12}
c) 2×10^{21}
d) 12×10^{12}

Q
An oncogene after mutation produces one of abnormal condition
Diabetes

a) Kidney stones
b) Encephalopathy
c) Encephalopathy
d) Cancer

Q
The cells which are activated initially by recognizing foreign antigen on APC

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b) T cells
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Q
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c) Point mutations
d) Chromosomal mutations

Q

Philadelphia chromosome is a chromosomal translocation between chromosome ___ and _____.

a) 9 & 22

b) 1 & 2

c) 7 & 21

d) 5 & 6

Q
Which one of the following gene is mutated in approximately 30% of all human tumors?

a) BRCA1 gene
b) EFGR gene
c) RAS gene
d) all of the above

Q
Genes are located in the nucleus and tangled charamatin material wrapped around histones in .....
a) Both Eukarotes and Prokaryotes
b) Eukaryotes
c) Prokaryotes
d) None of these

Q
Even in the absence of antigen humans can make more than..... different antibody molecules.
a) 10^{120}
b) 10^{12}
c) 12^{10}
d) 10^{112}

Q
Which one of the following helps in stimulating breast cells to produce milk?

a) STAT 1
b) Interferon
c) Erythropoietin
d) Prolactin

Q
Which one of the followings protein is activated when DNA is damaged,
a) Histone
b) Kinase
c) Actin

d) Albumin

Q
Which one of the following substances in immune system causes capillaries to dilate and become leaky?

a) Amino acids
b) Thymine
c) Histamine
d) Carbohydrates

Q
Some of the genes those who remain expressed all the time in cells are called?
a) Differentially expressed genes
b) Constitutively expressed genes
c) Housekeeping genes
d) Both housekeeping and constitutive genes

Q
a) Which one of the following substances in immune system causes capillaries to dilate and become leaky?
b) Thymine
c) Amino acids
d) Histamine
e) Carbohydrates

Q
Which one of the following helps in stimulating breast cells to produce milk?
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c) Interferon
d) STAT 1

Q
Which one of the followings protein is activated when DNA is damaged,
a) Actin
b) Kinase
c) Histone
d) Albumin

Q

What is the role of MHC in immunity? Enlist some major differences between MHC I and MHC II molecules.

Q
What is the homotypic and heterotypic fusion and how it forms?

Q
Differentiate Euchromatin and Heterochramatin

Q
What is the homotypic and heterotypic fusion and how it forms?

Q
Give the detail of events taking place at synapse at an activated neuromuscular junction?

Q
Differentiate between B cells and T cells?

Q
Please enlist the strategies of transcription regulation.....

Q
How UV light help to sterilization the equipment.....

Q

Please elaborate the different stages of the cell cycle.....

Q
How a sheep can be cloned through nuclear transfer, provide details in the form of steps?

Q
Describe the two signaling function of N-terminal ER signal sequence of a soluble protein.....

Q
How do terbutaline and β-blockers affect the function of Epinephrin adrenaline?

Q

Name five major growth promoting oncogenes.

Q
Whar are Stress Response Elements.....
.

Q
Please enlist the strategies of transcription regulation.....

Q
Give the detail of events taking place at synapse at an activated neuromuscular junction?

Q
Name five major growth promoting oncogenes.

Q
What are the Stem cells and how they differ from the specialized cells?

Q
Differentiate the "confluent monolayer" and "past confluency" stages.....

Q
Please elaborate the different stages of the cell cycle.....

Q
Please elaborate the different stages of the cell cycle.....

Q
Write detailed mechanism of TGF- β receptor family?

Q

How G protein is involved in the formation of GDP ultimately

Q

Write detailed mechanism of TGF- β receptor family?

Bio201

Why proteins denatur?
Down syndrome define?
Differentiate between purine and pyridine?
Describe the phases of mitosis?
What are the function of cellular membrane?
Translation?

BIO201

Q
……energy is the energy of motion
(a) Kinetic
(b) Potential
(c) Chemical
(d) Atomic
Q
Ribosomes can be free or attached with
(a) golgi apparatus
(b) lysosomes
(c) endoplasmic reticulum
(d) necleus
Q
Which organ/molecule can convert light energy
(a) NAD
(b) Retina
(c) Biotin
(d) Coenzyme A
Q
Beta pleated sheet is found in
(a) Primary protein structure
(b) Secondary protein structure
(c) Tertiary protein structure
(d) Quaternary protein structure
Q
Inhibitor that binds to a site other than active site of enzyme
(a) Competitive inhibitor
(b) Irreversible inhibitor
(c) Reversible inhibitor
(d) Non competitive inhibitor
Q
lipid act as a following in which many components of cell membrane floats
(a) Pond
(b) Lake
(c) River
(d) Stream

Q
How many macromolecules a cell has....
(a) 2
(b) 4
(c) 6
(d) 8
Q
Beta pleated sheet is found in
(a) Secondary protein structure
(b) Primary protein structure
(c) Tertiary protein structure
(d) Quaternary protein structure
Q
Cell membrane expels particles by vesicle formation
(a) pinocytosis
(b) Phagocytosis
(c) Endocytosis
(d) Exocytosis
Q
Molecule of phospholipids has ----------
(a) hydrophobic & Hydrophilic domain
(b) hydrophobic only
(c) hydrophilic only
(d) hydrogen linkage
Q
If one gene prevent the expression of other gene
(a) Pleiotropy
(b) Epistasis
(c) Synapse
(d) Tetrad
Q
Phenylalanine hydroxylase metabolizes
(a) Phenylalanine to Lysine
(b) Phenylalanine to Tyrosine
(c) Tyrosine to Phenylalanine
(d) Lysine to Phenylalanine
Q
Restriction enzymes inhibit the growth of
(a) Bacteriophage
(b) Bacteria
(c) Algae
(d) None
Q:
Missing of enzyme homogentisic acid oxidase caused
(a) Phenylketonuria
(b) Parkinson disease
(c) Alkaptonuria
(d) Cystic fibrosis
Q
Microtubules are originated from
(a) Cytoskeletal
(b) Nucleus
(c) Proteins

(d) Centrosomes
Q
The disintegration and condensation of nuclear material takes place in
(a) prophase II
(b) anaphase II
(c) metaphase II
(d) telophase II
Q
RNAs in large subunit of Eukaryotic ribosome
(a) 1
(b) 2
(c) 3
(d) 4
Q
Centromeres separate during whose anaphase:
(a) Mitosis
(b) meiosis I
(c) meiosis II
(d) None
Q
Enormous genetic information is present in
(a) RNA
(b) DNA.
(c) Nucleolus
(d) Nucleoplasm
Q
Okazaki fragments are joined by
(a) Ligase
(b) Gyrase
(c) Helicase
(d) Primase
Q
How much water present in a cell.....
(a) 50%
(b) 60%
(c) 70%
(d) 80%
Q
Charge on DNA is
(a) Positive
(b) Negative
(c) Neutral
(d) None
Q
Which of the following stabilizes cell shape
(a) DNA
(b) RNA
(c) Actin
(d) Myosin
Q
Define following terms?
(a) Endocytosis
(b) Phagocytosis

(c) Pinocytosis
(d) Exocytosis
Q
Which of the following stabilizes cell shape
(a) DNA
(b) RNA
(c) Actin
(d) Myosin
Q
……energy is the energy of motion
(a) Kinetic
(b) Potential
(c) Chemical
(d) Atomic
Q
Space between double membrane structure of endoplasmic reticulum is called
(a) Vesicles
(b) Leumen
(c) Cavity
(d) endoplasmic space
Q
----------is one of the example of co-factor
(a) Copper
(b) Biotin
(c) Haeme
(d) All
Q
lipid act as a following in which many components of cell membrane floats
(a) Pond
(b) Lake
(c) River
(d) Stream
Q
DNA replicates during.....
(a) G0 phase.
(b) G1 phase
(c) G2 phase
(d) S phase
Q
Difference between the alpha 1-4 glycosidic and beta 1-4 glycosidic bonds is on the basis of :
(a) Specificity
(b) Stability
(c) neutral pH
(d) size and shape
Q
Example of variable expressivity
(a) Dementia
(b) Huntington disease
(c) Alzheimer disease.
(d) Pancreatic necrosis
Q
Following has a major role in down syndrome
(a) Maternal age

(b) Fraternal age
(c) Child age
(d) Relative age
Q
DNA replication obeys
(a) Dispersive model
(b) Conservative model
(c) Semi-conservative model
(d) Distractive model
Q
Missing of enzyme homogentisic acid oxidase caused
(a) Phenylketonuria
(b) Parkinson disease
(c) Alkaptonuria
(d) Cystic fibrosis
Q
The RNA is --------- while DNA is ----------
(a) double stranded, double stranded
(b) single stranded, double stranded
(c) double stranded, single stranded
(d) partially double stranded, double stranded
Q
Ribosome after protein formation arrives at
(a) Lysosome
(b) Rough endoplasmic reticulum
(c) Smooth endoplasmic reticulum
(d) Golgi bodies
Q
The nucleus membrane is continuous with
(a) golgi apparatus
(b) ribosomes
(c) endoplasmic reticulum
(d) nucleus
Q
Power generating organelle of cell is:
(a) golgi bodies
(b) nucleus
(c) mitochondria
(d) cell membrane
Q
The disintegration and condensation of nuclear material takes place in
(a) prophase II
(b) anaphase II
(c) metaphase II
(d) telophase II
Q
An observable feature in the subject of genetics is called
(a) Trait
(b) Character
(c) Phenotype
(d) All
Q
Ribosome after protein formation arrives at

(a) Lysosome
(b) Rough endoplasmic reticulum

today my bio201
1: property of lipids (2)
2:diffrence purin & pyrimidine (2)
3:note on transcription (5)
4:function of lipids
or 2 que bhol gy hy , paper bht easy tha mera AlhamduLIlah bht acha hoa hy

tday's paper of mine.. bio201 (2:30)
techniques of genetic engineering (long),cell cycle(long), 3 dilemmas of DNA replication,mitochondrial features,restriction endonucleases,,macromolecules present in cell.

My Today's bio201 paper
Q1.diffentiale between dispersive, conservative , semi-conservative model of dna(5)
Q2.difference between Nucleoside vs Nucleotide?(3)
Q.3.difference between dna and rna?(5)
Q.4.Describe process of mitosis?(5)

bio201

differtiate between DNA and RNA
differtiate between nucleotide and nuceloside
Gel electrophoresis
eletronegativity
down syndrom
mitosis step

bio201 (cell biology) mid term papaer
Enlist the organelle of cell??
Function of DNA??
What is three dilemma of DNA replication??
Diff b/w cofactor, coenzyme and pathetic grp and its example??
what is the active site on an enzyme??

Q. ER single sequence is guided to the ER by which of the two components?

The ER signal sequence is guided to the ER by 2 components: 1) SRP which cycles between the ER membrane and the cytosol & binds to the signal sequence, & 2) SRP receptor in the ER membrane

Q. Write the name of antibody that appears on the surface of developing B cell?

IgM is the 1st class of Ab to appear on cell surface and secreted B cells can then switch Ab class

And IgD.

Q. What is the role of regulatory region explains its two elements.

Transcription Regulation:

• Some of genes are expressed in all cells all the time. Housekeeping genes are responsible for the routine metabolic functions (e.g. respiration). • Some are expressed as a cell enters a particular pathway of differentiation. • Some are expressed all the time in only those cells that have differentiated in a particular way. • Some are expressed only as conditions around and in the cell change. For example, the arrival of a hormone may turn on/off certain genes in that cell. Gene

Q. Describe the two signaling functions of N-terminal ER signal sequence of a soluble protein

The N-terminal ER signal sequence of a soluble protein has two signaling functions 1) It directs the protein to the ER membrane 2) It serves as a start-transfer signal (or start-transfer peptide) that opens the pore.

Q. What are the name of antibodies ?

5 classes of antibodies, IgA, IgD, IgE, IgG, & IgM, class of heavy chain—α, δ, ε, γ, & μ, respectively.

Q. Why Stem cells are different from specialized cells ?

Because Stem cells are unspecialized cells that renew themselves Under certain physiologic or experimental conditions, Stem cells can be induced to becodme cells with special functions.

Q. How UV helps in sterilization of equipment?

Sterilization, the killing of bacteria (or any types of cells), on medical instruments is primarily achieved by the radiation causing severe damage to the cell's components and to the cell's chromosomes, specifically the DNA. "Severe damage" to DNA is multiple breaks in the long DNA ladder-like structure. Radiation (for example, Ultra violet radiation) has enough energy to ionize atoms and molecules; that is, it can create charged particles and free radicals. In cells, the water can be ionized to form free radicals, somewhat like H-plus (H^+) and OH-minus (OH^-) radicals found in acids (for example H^+Cl^-) and bases (for example, Na^+OH^-). The free radicals made inside a cell's nucleus can cause damage to the cell's components and breaks in the DNA structure (the radiation may also directly cause ionizing breaks in

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the DNA's chemical chain). If enough damage is done to the cell's DNA, which acts as the cell's control mechanism, the cell cannot function properly or reproduce and the result is cell death (sterilization).

Q. What is stem cell and its benefits and therapeutic uses?

An undifferentiated cell of a multicellular organism which is capable of giving rise to indefinitely more cells of the same type, and from which certain other kinds of cell arise by differentiation.

Benefits

The hope going forward is that stem cells can also be used as a "renewable source of replacement cells and tissues" to treat common and serious diseases without the need for organ transplants or surgeries, including: macular degeneration, spinal cord injury, stroke, burns, heart disease, diabetes, osteoarthritis,

Therapeutic Uses

- Speed up the length of time it takes for injuries or wounds to heal
- Reduce pain, even **chronic joint pain**, with less need for medications
- Increase functionality, range of motion, flexibility and sleep quality
- Reduce muscle compensations and risk for future injuries
- Decrease nerve damage
- Increase collagen
- Help generate new heart and blood vessel tissue
- Help heal skin wounds, prevent formation of scar tissue and **reduce hair loss**
- Return patients to their normal activities as quickly as possible

Q. Describe three steps of RNA processing?

Pre-mRNA (freshly transcribed) is processed to make it functional.

1) Introns are removed during RNA maturation
2) The transcribed pre-mRNA is altered by the addition of a G cap modified GTP at the 5' end to protect RNA & facilitate translation.
3) After the last codon a poly A tail (100-300) is added at the 3' end for export and protection

Q. What is mechanism of the olfaction receptors..?

Humans can distinguish more than 10,000 distinct smells, which are detected by specialized olfactory receptor neurons in the lining of the nose. These cells recognize odors by means of specific G-protein-linked olfactory receptors, which are displayed on the surface of the modified cilia that extend from each cell. The receptors act through cyclic AMP. When stimulated by odorant binding, they activate an olfactory-specific G protein (known as Golf), which in turn activates adenylyl cyclase. The resulting increase in cyclic AMP opens cyclic-AMP-gated cation channels, thereby allowing an influx of Na+, which depolarizes the olfactory receptor neuron and initiates a nerve impulse that travels along its axon to the

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brain. It can detect thousands of different odor molecules through a large family of olfactory receptors (ORs), of a diverse protein sequence, which are located in olfactory sensory neurons (OSNs) in the olfactory epithelium in the nose of humans, and in the vomeronasal organ in animals.

Mechanism

Odorant molecules pass through the nasal cavity.

1: binding of odorant to its receptor activates G proten

2: the activation of G protein activates the synthesis of cAMP by adenylyl cyclase.

3: cAMP activates the opening site of ion channel

4: change in the ion concentration inside the cell send a signal to a specific area of the brain perceives the signal as a scent.

Q. Elaborate the mechanism of growth promoting mechanism of oncogenes?

Growth Promoting Oncogenes:

a) Growth Factors b) Growth Factor Receptors c) Signal-Transducing Proteins (secondary messengers) d) Nuclear Transcription Factors e) Cyclins and Cyclin Dependent Kinases (CDKs)

Autonomous Cell Growth In Cancer Cells Is Promoted By Following Elements

A) Growth Factor

Some cancer cells acquire the ability through mutations to produce growth factors to which they are responsive (autocrine signaling). Many glioblastomas secrete PDGF and sarcomas make TGF- α. Similar loops are fairly common many types of cancers.

B) Growth Factor Receptors

Mutated Growth Factor Receptors continue to deliver continues signals in the absence of the ligand. EGF receptor ERBB1 is over expressed in 80% of SCC of lung. ERBB2/HER2 is amplified in 25%-30% of breast cancers and adenocarcinomas of lung , ovary and salivary glands. These tumours are sensitive to small amounts of growth factors.

C) Signal-Transducing Proteins (Secondary Messengers)

Mutaion of RAS is the most common oncogene abnormality, approximately 30% of all human tumors contain mutated version of RAS gene. GAP/neurofibramin 1 (a RAS deactivating protein) is mutated in neurofibramatosis type 1. BCR-ABL hybrid gene activates RAS pathways (like p53 Normal ABL gene also promotes apoptosis via NLS domain).

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D) Nuclear Transcription Factors

MYC and the polymerase assemble on a specific DNA sequence and drive transcription of genes. MYC protooncogene is expressed in virtually all the cells, & the MYC protein is rapidly induced when cells receive a signal to divide. MYC binds DNA and causes transcriptional activation of many growth related proteins. Dysregulation of of MYC gene occurs in Burkitt Lymphoma (B-cell tumor); MYC is also amplified in many other cancers

Mutated Growth Factor Receptors continue to deliver continues signals in the absence of the ligand. EGF receptor ERBB1 is over expressed in 80% of SCC of lung. ERBB2/HER2 is amplified in 25%-30% of breast cancers and adenocarcinomas of lung , ovary and salivary glands. These tumours are sensitive to small amounts of growth factors.

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Q. How G protein involved in formation of GDP? 2 marks

When Rhodopsin absorbs the light (Photon) causing the G protein .transduction to exchange GTP for GDP.

Q. How sheep is cloned by the nuclear material. Explain it by all steps. 5 marks

Nuclear transfer is a form of cloning. The steps involve removing the DNA from an oocyte (unfertilised egg), and injecting the nucleus which contains the DNA to be cloned. In rare instances, the newly constructed cell will divide normally, replicating the new DNA while remaining in a pluripotent state. If the cloned cells are placed in the uterus of a female mammal, a cloned organism develops to term in rare instances. This is how Dolly the Sheep and many other species were cloned. Cows are commonly cloned to select those that have the best milk production.

Despite this, the low efficiency of the technique has prompted some researchers, notably Ian Wilmut, creator of Dolly the cloned sheep, to abandon it

Q. Write synapse and its steps involved in the mechanism. 5 marks

a junction between two nerve cells, consisting of a minute gap across which impulses pass by diffusion of a neurotransmitter.

Steps:

There are two general classes of neurotransmitters: large neuropeptides or smaller amines/amino acids which are synthesized differently.

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Step 1 – Neurotransmitter Synthesis. ...

Step 2 & 3- Neurotransmitter Packaging and Release. ...

Step 4 – Neurotransmitter Binding. ...

Step 5 – Stopping the Chemical Signal.

Mechanism

Neurotransmitters, such as Acetylcholine, are stored in vesicles in the presynaptic terminal. Acetylcholine is secreted across synaptic clefts, where it binds to the post synaptic membrane and causes a change in the permeability of the postsynaptic cell, which results in an excitatory post-synaptic potential

Q. Describe TGF-ß Receptor Family.

Consists of a large number of structurally related, secreted, dimeric proteins. During development, they regulate pattern formation and influence various cell behaviors, ie proliferation, differentiation, extracellular matrix production, and cell death. In adults, tissue repair and in immune regulation. Act through enzyme-linked receptors that are single-pass transmembrane proteins. Ligand first binds and activates a type-II receptor homodimer, which recruits, phosphorylates, and activates a type-I receptor homodimer, forming an active tetrameric receptor complex Activated TGF-β receptors phosphorylate Smad2 activated BMP receptors phosphorylate Smad1, Smad5, or Smad8.One of these Smads then dissociates from the receptor and binds to Smad4. Both type-I and typeII receptors, required for signaling. The type-I receptor directly binds and phosphorylates a latent gene regulatory protein of Smad family

Q. Define Electronegativity ?.

The tendency of an atom to attract electrons towards itself.

Q. What is Chargaff rule?

In DNA, the amount of Purines (A+G) is always equal to the amount of pyrimidine (T+C)

Q. Write mitochondrial features?

Mitochondria are double-membrane-enclosed organelle. It contains its own DNA, ribosomes, and other components required for protein synthesis, most of its proteins are encoded in the cell nucleus and imported from the cytosol There are two subcompartments in mitochondria: the internal matrix space and the intermembrane space Proteins imported into the matrix of mitochondria are usually taken up from the cytosol within seconds or minutes of their release from ribosomes. Mitochondrial proteins are first fully synthesized as precursor proteins in the cytosol & then translocated into mitochondria by a posttranslational mechanism.

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Q. Write Briefly note on cell cycle?

The cell cycle consists of four distinct phases: G_1 phase, S phase (synthesis), G_2 phase (collectively known as interphase) and M phase (mitosis). M phase is itself composed of two tightly coupled processes: karyokinesis, in which the cell's chromosomes are divided, and cytokinesis, in which the cell's cytoplasm divides forming two daughter cells. Activation of each phase is dependent on the proper progression and completion of the previous one. Cells that have temporarily or reversibly stopped dividing are said to have entered a state of quiescence called G_0 phase.

Interphase

Before a cell can enter cell division, it needs to take in nutrients. All of the preparations are done during interphase. Interphase is a series of changes that takes place in a newly formed cell and its nucleus, before it becomes capable of division again. It is also called preparatory phase or inter-mitosis. Previously it was called resting stage because there is no apparent activity related to cell division. Typically interphase lasts for at least 90% of the total time required for the cell cycle.

Interphase proceeds in three stages, G_1, S, and G_2, followed by the cycle of mitosis and cytokinesis. The cell's nuclear DNA contents are duplicated during S phase.

G_1 phase (growth)

The first phase within interphase, from the end of the previous M phase until the beginning of DNA synthesis, is called G_1 (G indicating *gap*). It is also called the growth phase. During this phase, the biosynthetic activities of the cell, which are considerably slowed down during M phase, resume at a high rate. The duration of G_1 is highly variable, even among different cells of the same species.[3] In this phase, the cell increases its supply of proteins, increases the number of organelles (such as mitochondria, ribosomes), and grows in size.

S phase (DNA replication)

The ensuing S phase starts when DNA synthesis commences; when it is complete, all of the chromosomes have been replicated, i.e., each chromosome has two (sister) chromatids. Thus, during this phase, the amount of DNA in the cell has effectively doubled, though the ploidy of the cell remains the same.

G_2 phase (growth)

G2 phase occurs after DNA replication and is a period of protein synthesis and rapid cell growth to prepare the cell for mitosis. During this phase microtubules begin to reorganize to form a spindle.

Mitotic phase (chromosome separation)

The relatively brief *M phase* consists of nuclear division (karyokinesis). It is a relatively short period of the cell cycle. M phase is complex and highly regulated. The sequence of events is divided into phases, corresponding to the completion of one set of activities and the start of the next. These phases are sequentially known as:

- prophase, metaphase, anaphase, telophase

Q. Define Genetic engineering?

Genetic engineering is the process of manually adding new DNA to an organism. The goal is to add one or more new traits that are not already found in that organism.

Q. What is Incomplete Dominance?

Incomplete dominance is a form of intermediate inheritance in which one allele for a specific trait is not completely expressed over its paired allele. This results in a third phenotype in which the expressed physical trait is a combination of the phenotypes of both alleles. Example 4 O'clock

Q. Define Codominance?

A situation, when both alleles appear in the phenotype neither allele is dominant. Both alleles are expressed in heterozygous individuals. Example Blood Group

Q. Define Mutation?

A **mutation** is a change that occurs in our DNA sequence, either due to mistakes when the DNA is copied or as the result of environmental factors such as UV light and cigarette smoke

Q. Define Epistasis?

Effect of one gene dependent on the presence of one or more 'modifier genes'

Q. What is Aquaporin?

Water molecules move through membranes by Osmosis (diffusion).

Q. What are PCR applications?

1) DNA Fingerprinting 2) Genotyping 3) Genomic Cloning 4) Gene expression studies 5) Drug discovery 6) classification of organisms etc.

Q. Write name of 5 antibiotics?

1) Tetra cycline 2) penicillin 3) amoxicillin 4) erythromycin 5) levofloxacin

Q. Define epinephrine and how it affected by beta blocker?

A hormone that is secreted by the adrenal gland in response to physical or mental stress, as from fear, and is regulated by the autonomic nervous system. The release of **epinephrine** causes an increase in heart rate, blood pressure, and respiratory rate. β-blockers Bind to the receptor but do not activate hormone induced effects.

Q. What are DNA replication Steps?

The three steps in the process of DNA replication are 1) initiation 2) elongation 3) termination.

Q. elaborates points between merged cAMP activity? 2 marks

cAMP-dependent protein kinases. Phosphorylate specific serine & threonine residues. Phosphorylation of many enzymes increases their catalytic activity, whereas phosphorylation of others decreases their activity.

Q. Define codon? 2 marks

A sequence of three nucleotides which together form a unit of genetic code in a DNA or RNA molecule called codon.

Q. Difference b/w klinefelter syndrome and turner syndrome.? 2 marks

Klinefelter syndrome is a genetic **disorder** that affects males. **Klinefelter syndrome** occurs when a boy is born with one or more extra X chromosomes (47 XXY)

Turner syndrome is a chromosomal condition in females that lack one X chromosomes they contain 45 chromosomes.

Q. How turbutalene and b blocker affect the function of epinephrine adrenaline.? 5 marks

β1- and β2-adrenergic receptors (Heart & Lungs Respectively) are coupled to G proteins (Gs), which activate adenylyl cyclase. α1 and α2 receptors are coupled to two other G proteins, Gq and Gi, respectively. adenylyl cyclase Inhibition & phospholipase C activation. Epinephrine bound to α2 receptors causes the arteries to constrict, cutting off circulation to the blood vessels in the intestinal tract, skin, and kidneys. (peripheral organs) Agonists (terbutaline) Mimic the function of a hormone by binding to its receptor and causing the normal response. Antagonists (β-blockers) Bind to the receptor but do not activate hormone induced effects. An antagonist acts as an inhibitor of the natural hormone (or agonist) by competing for binding sites on the receptor, thereby blocking the physiological activity of the hormone.

Q. Difference b/w confluent monolyer and post confluency stages?. 5 marks

A confluent monolayer is an adherent cell culture (dish, plate or flask) in which the cells have formed a single layer over the entire surface area available for growth. • Once the cells have started to form clusters above the first layer or have started to lift up from the surface, the culture is past confluency stage. • Cells that are repeatedly passaged at densities past confluency display decreased doubling times, decreased viabilities, & decreased ability to attach.

Q. Describe the larger subunit of ribosomes? 5 marks

T (transfer) site: first site where charged tRNA lands with the help of transfer factor A (AA) site: Anticodon binds to mRNA codon, P (polypeptide) site: tRNA adds AA to polypeptide, E (exit) site: after AA transfer tRNA exits here.

1) A charged tRNA with anticodon to next codon binds A site.

2) Large subunit breaks bond between tRNA & its AA at P site.

3) Assisted by elongation factors large subunit Catalyzes formation of peptide bond b/w the AA & the one attached to tRNA in A site (peptidyl transferase activity).

Q. Define carbohydrates and discuss it's classes? 3

Carbohydrates are polyhydroxy aldehyde or polyhydroxy ketones. It is further classified into 4 major groups.

The simplest **monosaccharides** are the two three-carbon trioses

Examples : – Glyceraldehyde,an aldotriose – Dihydroxyacetone, a ketotriose

Disaccharides are those sugars which yield two molecules of the same or different monosaccharides on hydrolysis – Maltose - 2 molecules of glucose on hydrolysis

Examples. Maltose , sucrose , galactose

Oligosaccharides are short chains of monosaccharide units or residues (310) joined by characteristic linkages called glycosidic linkage

Polysaccharides (Glycans) are those which yield more than 10 molecules of monosaccharides on hydrolysis • Some have hundreds or thousands of units • Some polysaccharides, such as cellulose, are linear chains; others, such as glycogen, are branched

Q. What do you know about endocytosis? Differentiation between phagocytosis and pinocytosis?

The material to be ingested is progressively enclosed by a small portion of the plasma membrane, which first invaginates and then pinches off to form an endocytic vesicle containing the ingested substance or particle

Phagocytosis involves the ingestion of large particles microorganisms or dead cells via large vesicles called phagosomes (>250 nm in diameter)

Pinocytosis involves ingestion of fluid & solutes via small pinocytic vesicles (about 100 nm in diameter)

Q. What are missense mutations? (2 marks)

Missense Mutations: Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein

Q. What are the types of homotypic and hetrotypic and how they formed?

After transport vesicles have budded from an ER exit site they shed their coat, and then they begin to fuse with one another this fusion of membranes from the same compartment is called **homotypic fusion**, to distinguish it from **heterotypic fusion**, in which a membrane from one compartment fuses with the membrane of a different compartment.

Q. What is TRANSMISSION OF SINGNAL FROM SYNAPSIS TO NEUROMUSCLE?

A **neuromuscular** junction (or myoneural junction) is a chemical **synapse** formed by the contact between a motor neuron and a muscle fiber. It is at the **neuromuscular** junction that a motor neuron is able to **transmit** a **signal** to the muscle fiber, causing muscle contraction.

Q. What is NEGATIVE and Positive CONTROL OF TRANSCRIPTION.?

The regulatory molecule (the repressor) prevents transcription. Negative Control of Transcription:

Positive Control of Transcription: the regulatory molecule (the CRP–cAMP complex) enhances transcription.

Q. Write ADULT STEM CELL AND ROLE?

Adult Stem cells Are harvest from there niches from a (postnatal) individual,No ethical dilemma for their use in research, Have a limited ability to renew themselves and differentiate in to different cell types. Do not cause an immune rejection.

Q. What is G proteins vision 2

Activates cyclic GMP phosphodiesterase in vertebrate rod photoreceptors.

Q. Write Role of MHC 2

The function of MHC molecules is to bind peptide fragments derived from pathogens and display them on the cell surface for recognition by the appropriate T cells.

Q. Difference of Transcription in prokaryotes & Eukaryotes3 marks

Prokaryotic Transcription	Eukaryotic Transcription
Occurs in the cytoplasm	Occurs in the Nucleus
There is no definite phase for its occurrence.	take place in the G1 and G2 phases of cell cycle
initiation of transcription does not need any proteins or initiation factors	initiation of transcription requires proteins called transcription factors like TFIIA, TFIIB , TATA BOX etc
A single RNA Polymerase synthesises all the three types of RNA(mRNA, tRNA, rRNA)	The RNA Polymerase I, II and III synthesizes rRNA, mRNA and tRNA respectively
The 23S, 16S and 5S rRNAs are formed from a single primary transcript	The 28S, 18S, 58S and 5S rRNAs are formed from two primary transcripts

Q. Write a note on cloning 5 marks

The term in itself means the *reproduction of a biological entity which is genetically identical to the one from which it originated.* The Greek term *'klon'* makes us think of a sprig from a plant which when inserted into mature soil can reproduce the plant from which it was taken.

Steps of Cloning

1) the reproduction of a cell line starting from a single cell: the cells produced are histologically identical to the ones from which they originated. Cloning is also spoken of for the reproduction of DNA fragments starting from a single DNA fragment;

2) the reproduction of embryos by disaggregating or subdividing an embryo (*embryo splitting*) in its early stages of development when the cells are totipotent or pluripotent, i.e., capable of developing into a complete organism. The embryos obtained are then transferred *in utero;*

3) the reproduction of genetically identical individuals through the *nucleo-transfer* of a somatic cell from an embryo, fetus or an adult individual, to a denucleated oocyte. The embryos obtained are later transferred *in utero*. The procedure can be repeated with several nuclei taken from the somatic cells of the same individual and inserted into denucleated ova;

4) the transfer of the nucleus of an oocyte to the cytoplasm of a second denucleated oocyte, as a form of "prevention" of mitochondrial diseases. This, however, is not cloning in a strict sense. The modified oocyte is later fertilized in vitro and transferred in utero.

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Q. Write Types of mutations 5 marks

Somatic mutations: Occur in somatic (body) cells & passed on to the daughter cells after mitosis but not passed on to the children.

Germ line mutations: Occurs in the germ line cells that give rise to gametes. A gamete with the mutation passes it on to embryo after fertilization.

Point mutations: mutations of single base pairs.

Chromosomal mutations: change the position or orientation of a DNA segment.

Silent Mutations: Mutations result in no change in amino acids due to redundancy genetic code

Nonsense mutations: Base substitution results in stop codon, (UAG) to form in the mRNA product

Missense Mutations: Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein

Frame-Shift Mutations: No substitutions. Single base inserted or deleted from DNA. Genetic message out of register "reading frame" changed

Q. What is Nuclear pore complex 10 marks

Eukaryotic nuclear envelope is perforated by large, elaborate structures called nuclear pore complexes which selectively facilitate nuclear transport. Each pore complex contains one or more open aqueous channels through which small water-soluble molecules can passively diffuse. Each complex has a molecular mass of about 125 million & is composed of aprox 50 different proteins, called nucleoporins, arranged octagonal symmetry. Small molecules (5000 daltons or less) diffuse in so fast that the nuclear envelope can be considered to be freely permeable to them, a protein of 17,000 daltons takes 2 minutes to proteins larger than 60,000 daltons are unable to enter the nucleus. Nuclear pore contains a pathway for free diffusion 9 nm in diameter & 15 nm long, this channel occupies a small fraction of total volume of the pore complex. The more active the nucleus is in transcription, the greater the number of pore complexes .The nuclear envelope of a typical mammalian cell contains 3000–4000 pore complexes. Cells synthesizing DNA, need to import about 106 histone molecules from the cytosol every 3 minutes to package DNA into chromatin, ie each pore complex needs to transport about 100 histone molecules/minute. Rapidly growing cell need to transport about 6 large & small ribosomal subunits/minute from nucleus, where they are produced, to the cytosol

Q. What is Complement system 10 marks

Complement System. Most of the time, the binding of antibodies to antigen performs no useful function until and unless it can activate an effector mechanism. Complement system serves several effector roles. Complement system provides the actual protection from the response while the interaction of antibodies and antigen provides the specificity of the response. Put another way, antibodies "finger" the target, complement destroys it the complement system consists of some 30 proteins circulating in blood plasma. These proteins are synthesized mainly in the liver, and they account for about 5% of the globulin fraction of blood serum. Most of these are inactive until .they is cleaved by a protease which, in turn, converts them into a protease. Thus many components of the system serve as the substrate of a prior component and then as an enzyme to activate a subsequent component. The classical pathway is

triggered by activation of the C1-complex (which consists of six molecule C1q and two molecules C1r and C1s), either by C1q's binding to antibodies from classes M and G, complexed with antigens, or by its binding C1q to the surface of the pathogen. C4b, which binds covalently to sugar residues on cell-surface glycoproteins,& a smaller, inactive, fragment of C4a which diffuses away C2 is cleaved into C2b, which binds non-covalently to a site on C4b, leaving a smaller, inactive, fragment of C2a which diffuses away

This binding leads to conformational changes in C1q molecule, which leads to the activation of two C1r (serine protease) molecules. Then they cleave C1s. (another serine protease) The C1-complex now binds to and splits C2 and C4, producing C2a and C4b

The complex of C4b•2b is called "C3 convertase" because it catalyzes the cleavage of C3 C3b binds covalently to glycoproteins scattered across the cell surface. Macrophages and neutrophils have receptors for C3b and can bind the C3b-coated cell or particle.

Q. Differentiate between pyrimidine and purines?

Purine is a double ring nitrogenous base while pyrimidine is a single ring nitrogenous base.

Purine contains two nitrogenous bases i.e Adenine and guanine while pyrimidine contains three nitrogenous bases i.e cytosine, thymine and uracil

Q. Enlist requirement for PCR?

1) Target DNA 2) Two oligonucleotide primers 3) DNA polymerase, 4) dNTPs 5) PCR machine

Q. DNA replication?

In molecular biology, DNA replication is the biological process of producing two identical replicas of DNA from one original DNA molecule. This process occurs in all living organisms and is the basis for biological inheritance. DNA is made up of a double helix of two complementary strands.

Q. Explain DNA replication Models?

Conservative replication: would preserve the original molecule and generate and entirely new molecule.

Semi conservative Replication: would produce molecules with both old and new DNA but each molecule would contain one complete old strand and one new one.

Dispersive Replication: would produce two molecules with old and new DNA interspersed along each strand

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Q. Function restriction endonucleases

Special bacterial enzyme can cut double stranded DNA at specific sites in a test tube. Restriction endonucleases (restriction enzymes) act as a kind of immune system, protecting the cell from the invasion of foreign DNA (virus).

Q. Difference between DNA and RNA? (5)

DNA	RNA
It is double stranded molecule	It is a single stranded molecule
Cytosine pairs with Thymine	Cytosine pairs with Uracil
It contains DeoxyRibose sugar	It contains Ribose sugar
It replicates to form new DNA	It cannot replicates itself
It is long lived molecule	It has short life span except a few RNAs

Q. Describe process of mitosis?(5)

Mitosis is a type of cell division in which one parent cell divides to form two daughter cells that contain same no of chromosomes this type of cell division occurs only in somatic cells. It has phases

1) Prophase: The beginning of mitosis. so the individual chromatids become visible. Microtubules form between each pole and the chromosomes to make up a spindle.
2) Prometaphase: is marked by the disappearance of the nuclear envelope. Chromosomes begin to move toward the poles.
3) Metaphase: All the centromeres arrive at the equatorial plate. Metaphase is the best time to see the sizes and shapes of chromosomes because they are maximally condensed.
4) Anaphase: Separation of the chromatids marks the beginning of anaphase, during which the two sister chromatids move to opposite ends of the spindle. Each chromatid con-tains one double-stranded DNA molecule and is now referred to as a daughter chromosome.
5) Telophase: Chromosomes stop moving at the end of anaphase, the cell enters telophase. The chromosomes begin to uncoil, the nuclear envelopes and nucleoli, reform their respective structures

Q. Gel electrophoresis

A gel is made up of agarose polymer suspended in a buffer. it sits in a chamber between two electrodes, depression in the gel (wells) are filled with DNA solution, restriction enzyme 1 cuts the DNA once resulting in fragments A and B. Restriction Enzyme 2 cuts the DNA once, at a different restriction sequences. if other restriction enzymes are used , two cuts are made in the DNA. Each sample is loaded into one well in the gel. as fragments of DNA move toward the positive electrode , shorter fragments move faster than longer fragments. The fragments can be identified by gel electrophoresis on the basis of their sizes.

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Q. Quaternary structure of protein

Protein **quaternary structure** is the number and arrangement of multiple folded protein subunits in a multi-subunit complex. It includes organizations from simple dimers to large homooligomers and complexes with defined or variable numbers of subunits.

Q. Define Tautomer

Tautomers are constitutional isomers of organic compounds that readily interconvert. The concept of tautomerizations is called tautomerism. The chemical reaction interconverting the two is called tautomerization.

Q 132 Write functions of fibriogen 2 marks

Fibrinogen, or factor I, is a blood plasma protein that's made in the liver.**Fibrinogen** is one of 13 coagulation factors responsible for normal blood clotting. When you start to bleed, your body initiates a process called the coagulation

Q. What is Alpaktonuria?

Alkaptonuria (black urine disease, black bone disease, or alcaptonuria) is a rare inherited genetic disorder in which the body cannot process the amino acids phenylalanine and tyrosine, which occur in protein.

Q. Write three stop codons?

1) UAA 2) UAG 3) UGA

Q. Write three RNA Viruses?

HCV , HBV , HIV , Adeno Viruses, INfluenza Virus , cytomegalo Virus etc

Q. How Different types of bacteria will produce

Binary fission ("division in half") is a kind of asexual reproduction. It is the most common form of reproduction in prokaryotes specially bacteria reproduce by binary fission.

Q. F. Griffith was used in his experiment? (2 marks

Griffith used two strains of Pneumococcus bacteria, type III-S (smooth) and type II-R (rough)

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Q. Differentiated between B-Cells and T-Cells

B- Lymphocytes (B-Cells)	T-Lymphocytes (T-cells):
1. They arise from bone marrow Bursa of Fabricus	1. They arise from bone marrow Thymus.
2. B-cells form humoral or antibody-mediated immune system (AMIS).	2. T-cells form cell-mediated immune system (CSM).
3. They defend against viruses and bacteria that enter the blood and lymph	They defend against pathogens including protists and fungi that enter the cells.
4. They are formed by the division plasma cells	4. They are formed by the division of lymphoblasts of three types: killer, helper and suppressor cells.
5. Plasma cells do not move to the site of infection.	5. Lymphoblasts move to the site of infection.
6. Plasma cells do not react against transplants and cancer cells.	6. Killer cells react against transplants and cancer cells.

Q. Define MHC and Differentiates between MHC-I and MHC-II. (10 Marks)

The function of MHC molecules is to bind peptide fragments derived from pathogens and display them on the cell surface for recognition by the appropriate T cells.

MHC I	MHC II
Present on the surface of all nucleated cells	Present on the surface of antigen presenting cells such as macrophages, B cells, dendritic cells etc.
Class I is a single chain (alpha chain) bound to an invariant partner	class II is a heterodimer containing two roughly identical chains (alpha and beta chains)
Class I present 8-10 amino acid peptides	class II presents 14-18 amino acid Peptides
Mainly interact with cytotoxic T cells	Mainly interact with helper T cells

Encoded by HLA-A, HLA-B and HLA-C genes	Encoded by HLA-D gene
Bind endogenous antigens synthesized in a cell	Binds exogenous antigens
Presence of foreign or over-abundant antigens targets cell for destruction	Presence of foreign antigens induces antibody production, and attracts immune cells to area of infection

Q. what is RFLPs and its function

What is RFLP: RFLP is a difference in homologous DNA sequences that can be detected as fragments of different lengths after digestion of the DNA samples.Most RFLP markers are heterozygous and highly locus-specific. RFLP probe is a labeled DNA sequence that hybridizes with one or more fragments of the digested DNA sample after they were separated by gel electrophoresis

Q. Difference between homozygous and heterozygous

A homozygous trait is when two of the same kind of alleles combines to form a trait. Eg. TT or tt

Heterozygous is a trait when two different kinds of alleles combine to form a trait. Eg Tt

Q. Difference between Stem cells and Specialized Cells

Stem cells differ from other kinds of cells in the body. All stem cells—regardless of their source—have three general properties: they are capable of dividing and renewing themselves for long periods; they are unspecialized; and they can give rise to specialized cell types

Groups of **specialized cells** cooperate to form a tissue, such as a muscle. Different tissues are in turn grouped together to form larger functional units, called organs. Each type of cell, tissue, and organ has a distinct structure and set of functions that serve the organism as a whole

Q. CaM kiase II role in movement of muscular tissue

Myosin light-chain phosphatase, more commonly called myosin phosphatase (EC 3.1.3.53), is an enzyme (specifically a serine/threonine-specific protein phosphatase) that dephosphorylates the regulatory light chain of myosin II. This dephosphorylation reaction occurs in smooth muscle tissue and initiates the relaxation process of the muscle cells. Thus, myosin phosphatase undoes the muscle contraction process initiated by myosin light-chain kinase. The enzyme is composed of three subunits: the catalytic region (protein phosphatase 1, or PP1), the myosin binding subunit (MYPT1), and a third subunit (M20) of unknown function. The catalytic region uses two manganese ions as catalysts to

dephosphorylate the light-chains on myosin, which causes a conformational change in the myosin and relaxes the muscle. The enzyme is highly conserved and is found in all organisms' smooth muscle tissue. While it is known that myosin phosphatase is regulated by rho-associated protein kinases, there is current debate about whether other molecules, such as arachidonic acid and cAMP, also regulate the enzyme.

Q. What is the role of rod cell in vision?

Rod cells are photoreceptor cells in the retina of the eye that can function in less intense light than the other type of visual photoreceptor, cone cells. Rods are usually found concentrated at the outer edges of the retina and are used in peripheral vision. Activation of photopigments by light sends a signal by hyperpolarizing the rod cell, leading to the rod cell not sending its neurotransmitter, which leads to the bipolar cell then releasing its transmitter at the bipolar-ganglion synapse and exciting the synapse.

Q. What is difference between Isotonic, Hypertonic and Hypotonic?

Isotonic	Hypertonic	Hypotonic
Isotonic solutions are two solutions that have the same concentration of a solute.	Hypertonic solution is one of two solutions that have a higher concentration of a solute.	Hypotonic solution is one of two solutions that have a lower concentration of a solute.

Q. Define Laminar flow cabinet principle and function?

Function: Provides sterile working area where cells can be handled without worrying about contamination.

Principal: In a **laminar flow hood** the air is passed through a HEPA (High Efficiency Particulates Air) filter which removes all airborne contamination to maintain **sterile** conditions. ... Now the **sterile** air **flows** into the working (flasking) area where you can do all your flasking work without risk of contamination.

Q. Write steps involving E-coli signaling

After ligand binding change in tertiary structure of extra cellular part of EnvZ leads to structural change in its cytoplasmic domain making it a kinase (auto..). EnvZ-P can now phosphor-ilate OmpR (responder) outside signal in and amplified. Receptor conformational change after ligand binding which activates kinase activity. Phosphorilation alters responder function. Signal amplified. Transcription factor activated. Protein synthesis results in altered cell activity.

Ligand binding causes a structural change permitting G protein to bind receptor. Binding of G protein to activated receptor causes it to exchange GDP for GTP (ligand releases).

Q. How antibody help the surface to produce B cells

Each B cell lymphocyte produces one unique antibody against one unique epitope. Once antibodies with sufficient specificity to the epitope can be encoded, the B cell begins to release antibodies into the bloodstream. These antibodies then bind specifically with the foreign molecule and allow the immune system to eliminate the molecule from the system.

Q. Define Gene linkage? (2)

Linkage is defined genetically as the failure of two genes to assort independently. Linkage occurs when two genes are close to each other on the same chromosome

Q. Define Palindromic Sequences?

Palindromic Sequences: A palindromic sequence is the same when read in 5′ to 3′ direction on either complementary strand of DNA. Example DENNIS SINNED GTAATG VS GTATAC

Q. VNTRs stand for?

Variable Number Tandem Repeats

Q. What are the steps of initiation of translation process?

a) **Shine Dalgarno sequence:**

In prokaryotes the rRNA of the small ribosomal subunit first binds to a complementary ribosome binding site upstream start codon AGGAGG this is called shine Dalgarno sequence on the mRNA.

b) Anticodon of methioninie charged to tRNA and bindes to start the codon (Methionine can be removed after translation).

c) In the small subunit rRNA stabilizes H-bonding between t and m RNA.

d) Large subunit facilitated by initiation factors binds the complex with methhionine charged tRNA at the P site.

Q. What are elongation steps in translation process?

a) A charged tRNA with anticodon to next codon binds A site.
b) Large subunit breaks bond between tRNA & its AA at P site.
c) Assisted by elongation factors large subunit Catalyzes formation of peptide bond between the AA and the one attached to tRNA in A site (peptidyl transferase activity).

Q. What are termination steps in translation process?

a) When stop codon enters A site elongation is terminates.

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b) Stop codons bind release factor, which hydrolysis bond between polypeptide chain and tRNA at P site.
c) Protein folding information is a function of AA sequence.

Q. Define Somatic mutations?

It Occur in somatic (body) cells & passed on to the daughter cells after mitosis but not passed on to the children.

Q. Define Germ line mutations?

It Occurs the germ line cells that give rise to gametes. A gamete with the mutation passes it on to embryo after fertilization.

Q. What are Point mutations?

These are Mutations of single base pairs.

Q. Write function of chromosomal mutations?

It change the position or orientation of a DNA segment.

Q. Define silent mutation?

The Mutations which results in no change in amino acids due to redundancy genetic code.

Q. Define Missense Mutations with example?

These are Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein. For example Sickle-cell disease results from a defect in human β-globin gene, a protein in human red blood cells that carries oxygen.

Q. Define Nonsense mutations?

Base substitution results in stop codon, (UAG) to form in the mRNA product

Q. Define Frame-Shift Mutations?

In which No substitutions accurs a Single base inserted or deleted from DNA. Genetic message out of register "reading frame" changed.

Q. What is epigenetics?

Epigenetics is the study of heritable changes in gene expression that do not involve changes in the DNA sequence.

Q. Define operon?

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Cluster of genes with 1 promoter is called operon Operon consists of a promoter, an operator, and two or more structural genes.

Q. Differentiate between Positive Control of Transcription and negative Control of Transcription?

Positive vontrol of Transcription is the trascription in which regulatory molecule (the CRP–cAMP complex) enhances transcription. While negative Control of Transcription In which The regulatory molecule (the repressor) prevents transcription.

Q. Write Three Systems of regulations of operon?

• An inducible operon regulated by a repressor protein

• A repressible operon regulated by a repressor protein

• An operon regulated by an activator protein

Q. Write two essential sequence of the promoter?

Promoter has two essential sequences. 1) Recognition sequence (RNA polymerase). 2) TATA box where DNA begins to denature.

Q. How pre-mRNA (freshly transcribed) is processed to make it functional?

1-Introns are removed during RNA maturation. 2) The transcribed pre-mRNA is altered by the addition of a G cap (modified GTP) at the 5′ end to protect RNA and facilitate translation. 3) After the last codon a poly A tail (100-300) is added at the 3′ end for export and protection.

Q. What is magnetism?

Magnetism can be a Signal which in some bacteria and birds orient themselves to Earth's magnetic poles.

Q. Define Signal Transduction?

The entire signaling process, from signal detection to final response is called signal transduction pathway/ signaling Cascades

Q. What are types of protein in signaling pathway?

1) Relay proteins: pass the message to the next signaling component in the chain.

2) Messenger proteins: carry the signal from one part of the cell to another (cytosol to nucleus).

3) Adaptor proteins: link one signaling protein to another, without themselves conveying a signal.

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4) Amplifier protein: either enzymes or ion channels, increase the signal they receive, (Produce large amounts of small intracellular mediators/activate signaling proteins (adenylyl cyclase).

5 Transducer proteins: convert the signal into a different form (adenylyl cyclase).Gnaling.

6 Bifurcation proteins: spread signal from one pathway to another.

7 Integrator proteins: receive signals from 2++ pathways integrate them before relaying a signal onward.

8 Latent gene regulatory proteins: activated at the cell surface then migrate to nucleus to stimulate gene transcription.

Q. Write at least one function of scaffolds protein?

Many cells use scaffold proteins to enhance speed efficiency & specificity of response.

Q. What is abbreviation of GPK, GP, PP, GS?

glycogen phosphorylase kinase (GPK), glycogen phosphorylase (GP) phosphoprotein phosphatase (PP), glycogen synthase (GS)

Q. What is difficult between Agonists and antagonists?

Agonist (terbutaline) Mimic the function of a hormone by binding to its receptor and causing the normal response while Antagonist (β-blockers) Bind to the receptor but do not activate hormone induced effects.

Q. What are the causes of cAMP?

cAMP rise causes membrane proteins to permit a massive flow of water from the blood into the intestinal lumen, causing diarrhea (water loss) resulting in death from dehydration.

Q. How CML formed?

chronic myelogenous leukemia (CML) results from a chromosomal translocation between chromosome 9 and 22.

Q. What Is Bruton's tyrosine kinase,?

Bruton's tyrosine kinase, a protein whose mutation is associated with X-linked agammaglobulinemia.

Q. What is the function of Erythropoietin?

Erythropoietin stimulates erythrocyte precursor cells to survive, proliferate, and differentiate, is associated with Jak2

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Q. Write function of prolactin?

Prolactin stimulates breast cells to produce milk Activated STAT5 stimulates the transcription of genes that encode milk proteins.

Q. What do you know about TGF-β Receptor Family?

Consists of a large number of structurally related, secreted, dimeric proteins. During development, they regulate pattern formation and influence various cell behaviors, ie proliferation, differentiation, extracellular matrix production, and cell death. In adults, tissue repair and in immune regulation. Act through enzyme-linked receptors that are single-pass transmembrane proteins.

Q. How many receptors are required for signaling mentioned their name?

Two types of receptors are required for signaling Both type-I and type-II receptors, required for signaling.

Q. What is abbreviation of PDGF and EGF?

platelet-derived growth factor (PDGF) and epidermal

growth factor (EGF).

Q. Write characteristics of Jak stat signaling?

Jak-STAT Signaling.Increase antigen presentation of macrophages.Activate & increase lysosome activity macrophages.Cells to increase expression of MHC class I .Promotes adhesion and binding required for leukocyte migration.Promotes NK cell activity.Activate inducible Nitric Oxide Synthase.Jak-STAT one of the most direct signaling.Ligand binding induces receptor to dimerize.Binding brings Jaks close to cross-phosphorylate each other, increasing kinase activity. The Jaks then phosphorylate receptors, creating STAT docking sites

Q. Define antigens?

Substances that activate adaptive immune specific (AIS) are called antigens.

Q. Write names of five class of antibodies?

5 classes of antibodies, IgA, IgD, IgE, IgG, & IgM,

Q. Explain immunological memory?

Immunological Memory: The adaptive immune system, like the nervous system, can remember prior experiences. First exposure to an antigen A: Immune response appears after several days, rises rapidly and exponentially, and then, more gradually, declines. Secondary immune response: Lag period is shorter, and the response is greater indicating animal remembers its first exposure to antigen A. If

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animal is given a different antigen B instead of second injection of antigen A the response is typical primary not secondary, immune response.

Q. What are Cellular Basis of Immunological Memory?

When naïve lymphocytes are stimulated by their specific antigen, they proliferate and differentiate. Most become effector cells which function & then die, while others become long-lived memory cells.

Q. What is mechanism of class switching?

Class Switching: B-Cell initially produces IgM (antibody's heavy chain is encoded by the first constant region gene, the μ segment) molecules, which recognise specific antigenic determinant. When B cell becomes a plasma cell another deletion occurs in the cell's DNA, which position heavy-chain variable region gene next to a different constant region (γ, ε, or A)

Q. What are the types of MHC molecules?

There are two types of MHC molecules MHC class I and MHC class II with distinct functions.

Q. What is genetic diversity of MHC classes?

3 genes for class I proteins (HLA-A, B &C). 3 genes for class II proteins (HLA-DP, DQ &DR)6 genes each for both the classes.

Q. What happened after T cell activation?

Once a helper T cell is activated by an antigen-presenting cell to become an effector cell, it then help activate other cells.

Q. what is TH1 and TH2 explain them?

When activated by an antigen-presenting cell, a naïve helper T cell can differentiate into either of two distinct types of effectorhelper cell, called TH1 and TH2. TH1 cells mainly help activate macrophages and cytotoxic T cells, whereas TH2 cells mainly help activate B cells.

Q. Explain complement system?

Complement System: Most of the time, the binding of antibodies to antigen performs no useful function until and unless it can activate an effector mechanism.Complement system serves several effector roles.Complement system provides the actual protection from the response while the interaction of antibodies and antigen provides the specificity of the response.Put another way, antibodies "finger" the target, complement destroys it The complement system consists of some 30 proteins circulating in blood plasma.These proteins are synthesized mainly in the liver, and they account for about 5% of the globulin fraction of blood serum.Most of these are inactive until .they are cleaved by a protease which,

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in turn, converts them into a protease.Thusmany components of the system serve as the substrate of a prior component and then as an enzyme to activate a subsequent component.

Q. What do you know about proto oncogene? And write its classes?

A proto-oncogene is a normal gene that can become an oncogene, either after mutation or increased expression. Proto-oncogenes code for proteins that help to regulate cell growth& differentiation.

Classes : Growth Promoting Oncogenes, Growth-Inhibiting Cancer Suppressor Genes, DNA Repairing Genes, Apoptosis Genes.

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BIO-201 CELL BIOLOGY MCQS FOR FINAL TERM

1 : The small sub unit of Ribosomes is bind to mRNA this sequence is called ?

A) Plaindromic sequences B) Shine sequence C) Shine –Dalgarno sequence D) Dalgarno sequence

2 : After translation,.....can be removed?

A) Alanine B) Methionine C) Arginine D) Lysin

3 : rRNA in small subunit stabilizes ..... between m and t RNA?

A) H –bonding B) C – bonding C) I – bonding D) M – bonding

4 : Large subunit breaks bond between .....?

A) mRNA B) rRNA C) SNRNa D) tRNA

5 : Protein folding information is a function of..... Sequences?

A) UU B) AA C) GG D) CC

6 : Stop codon is....?

A) AUG B) UCA C) UGA D) UAG

7 : Protein sequences begins on.....?

A) cytoplasm B) Nucleus C) Free ribosomes D) Ribosomes

8 : The signal sequences for the action of protein is present in.....?

A) Polypeptide chain B) Smooth ER C) A site D) P site

9 : An NLS is necessary for import of protein into the.....?

A) Cytoplasm B) Nucleus C) Prokaryotes D) Eukaryotes

10 : Which mutation cells passed on to the daughter cells after mitosis?

A) chromosomal mutation B) Point mutation C) Somatic mutation D) Germ line mutation

11 : Point mutation has base pair?

A) Double B) Single C) Triple D) None

12 : No change in amino acid sequences in.....?

A) Point mutation B) Somatic mutation C) Silent mutation D) Chromosomal mutation

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13 : A protein that carries oxygen in human.....?

A) White blood cells B) Bone cells C) Red blood cells

14 : .....is the study of heritable changes in gene expression?

A) Epistasis B) Epigenetics C) Genetics D) Genome

15 : Cluster of genes with .....promoter is called operon?

A) 1 B) 2 C)3 D) 4

16 : Which polymerase binds to initiate transcription of lac operon?

A) DNA B) RNA C) mRNA D) tRNA

17 : Transcription is blocked by .....process?

A) 1 B) 2 C) 3 D) 4

18 : Tryptophan blocks RNA polymerase from.....?

A) Binding B) Transcribing genes C) DNA polymerase D) Binding and transcribing genes

19 : Which regulatory molecule enhances transcription?

A) CRP – cAMP B) CRP C) cAMP D) None

20 : An operon regulated by.....?

A)Repressor protein B)Activated protein C) Structural genes D) Regulatory protein

21 : Genes are located in nucleus in a.....material?

A) Nucleolus B) Chromatin C) Tangled chromatin D) Cytoplasm

22 : DNA is highly ordered around positively charged .....?

A) Repressor proteins B) Histone proteins C) Activated proteins D) Regulatory proteins

23 : Which groups reduce positive charge when add to histones?

A) Alkyl B) Carboxy C) Acetyl D)Ketone

24 : Promoter has .....essential sequences?

A) One B) Two C) Three D) Four

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25 : TATA box where ……..begins to denature?

A) RNA B) mRNA C) DNA D) rRNA

26 : Eukaryotic promoter have TATA box ,which is located…….?

A)25bp B)26bp C)27bp D)28bp

27: The first transcription factor is…….?

A) TFD11 B) TF11D C) T11FD D) FT11D

28 : Many genes have enhancer away ……..nucleoitide?

A) 1000 B) 2000 C) 3000 D) 4000

29 : During RNA maturation introns are…….?

A) Produced B) Removed C) Increase D) Decrease

30 :RNA has complimentary bases that bind CS at…….exon – introns boundery?

A) 3 ’ B) 5’ C) 3’- 5’ D) 5’-3’

31 :The active X makes RNAi anti –Xist gene appropriately called…?

A) Tsix B) Tseven C) Teight D) Tnine

32 Which attract chromosomal proteins?

A) Methylation B) Histone deacetylation C) a and b D) None

33 : Multicellular organism cells are specialized for the support ……?

A)Table sugar B) Sweat sugar C) Brain sugar D) All of these

34 :Our sense organ allow us to chemical respond to…..?

A) Odor B) Light c)Taste D) Touch

35 : Signal detection to final response is called…..?

A) Signal translation pathway B) Signal transduction pathway
C) Signal transcription pathway D) Signal transformation pathway

36 :Which signals receptors on the cells that secret them?

A)Paracrine signals B)Transdution signal C)Autocrine signal D)Translation signal

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37 :Cell behavior in an environment flooded with.....of ligands?

A) Hundred B)Thousand C)Millions D)Twenty

38 : Inactive ICR may be.....bond?

A)RNA B)mRNA C)rRNA D)DNA

39 : Activated receptor binds DNA induces gene.....?

A)Transdution B) Transcription C)Translation D)Transformation

40 :How many types of proteins?

A) 5 B) 6 C) 7 D)8

41 : Which protein spread signal from one pathway to another?

A) Bifurcation B) Integrator C) Amplifier D) Transducer

42 : Which protein produce intracellular mediators?

A) Relay B) Amplifier C) Adopter D) Gene regulatory

43 :Many cells use .....proteins to enhance protein?

A) Transducer B) Amplifier C) Relay D) Scaffold

44 :.....are produced by in response to signal received by the receptor?

A) SCIMs B)SICMs C)SMICs D)SMCIs

45 : EnvZ change the cytoplasmic domain and make it a.....?

A)Ligase B)Helicase C)Kinase D)primase

46 : Helices .....that participate in agonist binding?

A)3 , 5 and 6 B)3 , 4 and 6 C) 3 , 4 and 7 D) 3 ,4 and 8

47 :Ligand activates .....protein?

A)T B) C C) G D) A

48 :cAMP can bind .....to open them?

A) A protein B) G protein C) Ion channels D) Ionic bond

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49 : One receptor ..... active Gs?

A) 100 B)200 C)300 D)400

50 : Per receptor –hormone complex are produced by .....cAMP?

A) Millions B)Thousand C) several thousand D)Hundred

51 : Release of glucose fuels.....response?

A)Fight B) Flight C) Fight and flight D) Fight or flight

52 : AKA cAMP depend protein.....?

A) Kinase B) Ligase C) Helicase D) Primase

53 : The PKAs are.....?

A)Hexamerse B) Hetramerse C) pentamerse D) Tetramerse

54 : PDE1 receptor interact with.....?

A) G_{ia-} B) G_{BY-} C) G_{Y-} D) G_i

55 : G_{BY} open K channel in .....?

A) Blood cells B) Brain cells C) Bone cells D) a and c

56 : The G protein the synthesis of cAMP by.....?

A) Adenylyl cyclase B) Adinosine phosphate C) G_i D) cyclase

57 : cAMP causing diarrhea resulting in death from .....?

A) Hydration B) Dehydration C) Duffution D) osmosis

58 : One rhodopsin molecule absorbs .....proton?

A) One B) Two C) Three D) Four

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59 : Calmodulin binds 4 Ca^{2+} allosteric (PKA).....?

A) 1% B) 2% C) 3% D) 4%

60 : Tired brain produces.....?

A) Cytosine B) Guanine C) Adenosine D) Thymine

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61 : Extracellular ligand – binding domainon .....segment?

A) 1 transmembran B) 2 trnsmembran C) 3 trnsmembran D) 4 trnsmembran

62 : Most numerous receptor tyrosine.....?

A) ligases B) Helicases C) Kinases D) primases

63 : EGF activated MAPK stays active for .....mint?

A) 8 B) 7 C) 6 D) 5

64 :.....genes encode kinases?

A) 2 % B) 3 % C) 4% D) 5 %

65 : .....proteins are phosphorilated ?

A) ¼ B) 1/5 C) 1/3 D) 1/6

66 : Acitvated Raf is phosphorylates many molecules of.....?

A)MEK B) MAP C) MKE D) MPA

67 : Chromosomal translocation between chromosome.....?

A) 9 and 52 B) 9 and 42 C) 9 and 32 D) 9 and 22

68 : APC gene mutations accur in human colon cancer.....?

(a). 80% (b). 90% (c). 70% (d). None of these

69 : Proteolysis means: protein's .....?

(a). Breaking (b). Production (c). None of these (d). All of these

70 : APC adherens junctions scaffold protein called.....?

(a). Axin (b). Arginin (C). Both (d). None

71: Stimulation of milk production is due to.....?

(a). Erythropoientin (b). Prolaction (c). Growth hormone (d). Alpha. Interferon

72 : ABL protein preventing ATP from.....?

A) Produced B) Removed **C) Binding** **D) CML**

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73 : BTK and PLC – Y to the cytoplasmic face of the.....?

A) Nucleus B) Cell membrane C) plasma membrane D) All of these

74 : Bruton,s tyrosine mutation is associated with .....agammaglubulemia?

A) Y – linked B) X – linked C) XY – linked D) XX – linked

75 : Which signal cells indirectly activating protein kinase B?

A) PDK1 B) PKB C) PI 3 D) PIP3

76: Intracellular protease cleaves off the cytoplasmic tail of _____ when it binds _____.?

(a). Notch, delta (b). Delta, notch (c). Notch, notch (d). Delta, delta

77 : Activated protein is .....?

A) PDK1 B) PKB C) PI 3 D) PIP 3

78 : Which Recepter required for signalling:

(a). Type_1 TGF.beta (b). Type_2 TGF.beta (c). Both D) None of these

79: Smad complex move into_____ for association of other gene regulatory proteins:

(a). Nucleus (b). DNA (c). Target Gene (d). All of these

80 : _____ protein called Dishevelled is activated when Wnt is present.

(a). Nucleus signalling (b). Cytoplasmic signaling (c). Contact dependent (d) None

28 june : lec 119 – 128

81 : Which is the primary concerns due to skin is lost?

A) Infectious particals B) Plasmodium infection C) Injury infection D) None

82 : Pathogen may cause infection of.....?

A) Digestive system B) Immune system C) Nerve system D) All of these

83 : Our tears and saliva have enzymes that destroy .....?

A) Organisms B) Macroorganisms C) Microorganisms D) None

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84 : How many steps of immune system?

A) 7 B) 6 C) 5 D) 4

85 : Histamine diffuse into .....?

A) Vains B) Artries C) Capillaries D) All of these

86 : Phagocytes engulf .....and dead cells?

A) Bacteria B) Plasma C) Enzyme D) Histamine

87 : Erythrocytes transport oxygen and.....?

A) Carbohydrate B) Protein C) Carbon dioxide D) None

88 : Which cells differentiate form antibody- producing cells and memory cells ?

A) Dendritic cells B) Mast cells C) T cells D) B cells

89 : Which cells release histamine when damaged?

A) Mast cells B) Basophils C) Neutrophils D) T cells

90 : T cells mature in .....?

A) Lymphatic vessel B) Medulla C) Thymus D) Bone marrow

91 : Lymph nodes contain .....?

A) REDs B) Tissues C) WBCs D) All of these

92 : AIS has.... Component?

A) Two B) Three C) Four D) Five

93 : B lymphocyte kill pathogen.....?

A) 20% – 60% B) 20% - 50% C) 20% - 40% D) 20% - 30%

94 : How many lymphocytes in human body?

A) 2×10^{2} B) 2×10^{12} C) 2×10^{22} D) 2×10^{32}

95 : Substance that activate AIS are called?

A) Antigens B) Antibody C) Antibody protein D) AIR

96 : The rates were unable to mount.....?

A) AIS B) IIS C) AIR D) None

97 : B cells develop in.....?

A) Thymus B) peripheral lymphocyte organ C) Bone marrow D) Hemopoietic tissue

98 : The B – cells is composed of four polypeptide chains , these are....?

A) 1 identical heavy chain and 3 identical light chains B) 2 identical heavy chains and 2 identical light chains C) 3 identical heavy chains and 1 identical light chains D) 4 identical heavy chains

99 : Immunoglobin domain is.....long?

A) 110 AA B) 111 AA C) 112 AA D) 113 AA

100 : Polyvalent.....times stronger than monovalant ?

A) 400 B) 300 C) 200 D) 100

101 : Which is the first class of Ab to appear on cell surface?

A) IgE B) IgM C) IgA D) IgD

102 : How many classes of antibodies?

A) 5 B) 10 C) 12 D) 15

103 : Which chain is required for pentamer formation?

A) L chain B) J chain C) H chain D) None

104 : IgG is monomer ,it is only.....?

A) IIS B) AIR C) AIS D) Ab

105 : Which receptors are located on the mast cells?

A) IgA Fc B) IgE Fc C) IgM Fc D) IgG Fc

106 : A myeloma cell is fused with a B cell to form.....?

A) Spleen B) Native cells C) Beta cells D) Hybridoma

107 : In the absence of antigen human can make more than .....different antibody molecule?

A) 10^{12} B) 10^{13} C) 10^{14} D) 10^{15}

108 : Number of human gene is.....?

A) 30,000 B) 40,000 C) 50,000 D) 60,000

109 : Immunoglobulin genes have relatively.....mutation rate?

A) Low B) High C) a and b D) very low

Mcqs 129 – 138

110) Effecter helper T cell stimulate the responses of other cells.....?

A) Phagocitic macrophages , B cells B) macrophages , T cells
C) Intracellular phathogen D) Cytotoxic T cells

111) T cells receptor recognizes processed antigen bond to a class .....?

A) T cells receptor B) MHC protein C) BCR gene D) All of these

112) How many protein are not immunoglobulins ?

A) 3 B) 2 C) 1 D) 4

113 : Antigen is .....?

A) Lymphoid organ B) Dendritic cell C) Protein antigen D) Peptide fragment

114) A virus infected cell in the case of .....?

A) Cytotoxic C cell B) Effecter helper T cell C) MHC genes D) None

115) How many types of MHC molecules?

A) 1 B) 2 C) 3 D) 4

11 6) APC express major histocompatibility compex.....?

A) cytotoxic T cell B) Macrophages b cells C) MHC genes D) Protein antigen

117) How many genes are present in MHC classes ?

A) 3 B) 4 C) 5 D) 6

118) 3 genes for class 1 proteins.....?

A) HLA – A , B , C B) HLA – A , C , D C) HLA – DP , DQ D) HLA – DQ , DP

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119) Class 11 can .....pair up?

A) α and γ B) α and β C) α , β , γ D) None

120) MHC groove can accommodate an extended peptide about .....long?

A) 8 – 30 amino acids B) 8 – 20 amino acids C) 8 – 10 amino acids D) 8 – 11 amino acids

121) Each class 1 MHC protein can bind a peptide of....?

A) Parallel sequences B) Perpandicular sequences C) Multiple sequences D) Diverse sequences

122) Which terminal bond to invariant pockets ?

A) C B) N C) C , N D) C , N , P

123) Cytotoxic T cells provide protection against .....?

A) Protein antigen B) Macrophage pathogen T cell C) MHC genes D) Intracellular pathogen

124) Which cell form a clone?

A) T_c B) T_b C) T_a D) T_d

125) How many steps is present in perforin – depending killing?

A) 3 B) 4 C) 2 D) 1

126) Helper T cell is activated by an .....cell to become an affector cell?

A) Macrophage pathogen cell B) MHC C) Antigen cell D) All of these

127) How many types of effector cell ?

A) 5 B) 4 C) 3 D) 2

128) TH2 cells mainly help activate .....?

A) B cell B) T cell C) B , T D) None

129) APCs provide .....types of signal?

A) One B) Two C) Three D) Four

130) Signal 1 is provided by.....?

A) B 7 protein B) MHC protein C) Peptide bond D) Costimulatory protein

131) B 7 protein are ……..?

A) CD80 – CD 83 B) CD80 – CD 84 C) CD80 – CD 85 D) CD80 – CD 86

132) Which protein is present on the surface of T cell?

A) CD 28 B) CD 29 C) CD 30 D) CD 31

133) The compliment system consist of ……..protein?

A) 40 B) 30 C) 20 D) 10

134) The compliment protein are account for about …….of the globulin?

A) 3 % B) 4 % C) 5 % D) 6 %

135) C1 compex consist of …..molecules?

A) Three B) Four C) Five D) Six

136) Which molecule binding the surface of pathogen?

A) C1q B) C1r C) C1s D) All of these

137) The conformational changes leads to C1q to the activation of two…..?

A) C1s B) C1r C) C4 D) C2a

138) The complex of C4b . 2b is called?

A) C1 B) C2 C) C3 D) C4

139) Marcophages and neutrophils have receptor for….?

A) C1b B) C2b C) C3b D) C4b

140) Cleavage of C5 by the …?

A) C1b / C5 B) C1 / C5 C) C3b / C5 D) C3 / C

Topic : 139 – 148

141) C3b binds a protein called?

A) Factor B B) Factor C C) Factor D D) Factor E

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142) C3 convertase into .....form?

A) 2 B) 3 C) 4 D) 5

143) The MBL – associated serine proteases,.....?

A) MASP-2 , MASP- 4 ,MASP- 6 B) MASP -3 ,MASP-5,MASP-7

C) MASP-1,MASP-2,MASP-3 D) All of these

144) The explosive potential of the compliment system requires at least .....protein?

A) 9 B) 10 C) 11 D) 12

145) Factor 1 is inactivates.....?

A) C3b B) C5 C) C3 D) Bb

146) C1 inhibitor binds to sites on activated.....?

A) C 1 and C 2 B) C 1r and C 2s C) C 2 and C 4 D) C 2r and C 4s

147) C 4 and C 2 deactivated by .....?

A) CNH11 B) CHN11 C) C11NH D) C12NH

148) A proto – oncogene can become a oncogene after.....?

A) Translation B) Transcription C) Transformation D) Mutation

149) Proto – oncogene code for .....?

A) Protein B) Enzyme C) Transformation D) Translation

150) Proto – oncogene are involved in signal .....?

A) Transformation B) Transdution C) Translation D) Transcription

151) How many classes of oncogenes?

A) 2 B) 3 C) 4 D) 5

152) Autonomous cell growth is present in.....cell?

A) Tumors B) Virus C) a and b D) Cancer

153) Many glioblastomas secrete .....?

A) PDGF B) PGDF C) PFGD D) PFDG

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154) EGF receptor ERBB1 is over expressed in .....of SCC of lung?

A) 70 % B) 80 % C) 90 % D) 100 %

155) ERBB2 / HER2 is amplified in .....of cancer?

A) 25 % - 60 % B) 25 % - 70 % C) 25 % - 40 % D) 25 % - 30 %

156) Mutation of RAS is present approximately .....in all human tumors?

A) 30 % B) 40 % C) 50 % D) 60 %

157) BCR – ABL hybrid gene activates.....?

A) MYC B) RAC C) MAP D) ERBB2

158) MYC assemble on a specific DNA sequence and drive .....of genes?

A) Translation B) Transformation C) Transcription D) Transduction

159) MYC binds with.....?

A) RNA B) mRNA C) Protein D) DNA

160) MYC gene encoding the .....?

A) Cyclin D B) Cyclin E C) Cyclin F D) Cyclin G

161) RB which overcome the hurdle of .....hurdle?

A) G1 to G2 B) G1 to S C) G1 to M D) All of these

162) The activated of CDKs is regulated by .....?

A) CDK1 B) CDK2 C) CDK D) CDK4

163) CDKS is occur in .....?

A) TGF – γ B) TGF – β C) TGF – γ ,β D) TGF – α

164) RB bind factor E2F when RB is phosphorilated by.....?

A) CDK4 , CDK6 , CDK2 B) CDK2 , CDK3 , CDK4 C) CDK5 , CDK 6 , CDK7 D) All of these

165) Which can activate DNA repair protein?

A) TP53 B) p21 C) P53 D) CIP

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166) Which is most commonly mutant gene in human cancers?

A) p21 B) CIP C) TP53 D) P53

167) P53 protein is stabilize by.....?

A) Damaged DNA B) Undamaged DNA C) Damaged RNA D) Undamaged RNA

168) How many types of Cdks?

A) 5 B) 4 C) 3 D) 2

169) Active p53 binds to requlatory region of .....gene?

A) p20 B) p21 C) p23 D) p24

170) Abnormally high level of Myc cause the activation of .....?

A) PARF19 B) PRAF19 C) P19ARF D) PA19RF

171) One of the p53 alleles mutated which causes .....fold increased chance of developing malignant tumor by age.....?

A) 25 , 25 B) 25 , 35 C) 25 , 45 D) 25 , 50

172) Papillomavirus uses .....viral proteins?

A) Two B) Three C) Four D) Five

173) Two vital proteins are.....?

A) E5 .E6 B) E6 . E7 C) E7 .E8 D) E8 .E9

174) Which virus uses a single dual purpose protein?

A) Rb B) Mdm2 C) Sv40 D) p53

175) How many bone cells die in a healthy adult?

A) Hundred B) Thousand C) Millions D) Billions

176) .....biochemical processes take place in membrane?

A) Vital B) Viral C) Protein D) None

177) An animal cell contains about.....protein molecule?

A) 10^{9} B) 10^{10} C) 10^{11} D) 10^{12}

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178) How many kinds of protein molecule?

A) 10,000 – 40,000 B) 10,000 – 30,000 C)10,000 – 20,000 D)10,000 – 10,000

179) Lipid bilayer of organelle membranes is impermeable to most.....molecule?

A) Hydrophobic B) Hydrophilic C) a and b D) None

180) The transport vesicles molecule derived from the lumen of .....compartment?

A) One B) Two C) Three D) Four

181) .....direct protein from the cytosol into the ER?

A) SS2 B) SS1 C) a and b D) Mdm2

182) SS2 consist of a specific .....dimensional arrangement of atoms on the protein surface ?

A) One B) Two C) Three D) Four

Mcqs 149 – 158

183) Envelop consists of ....concentric membrane?

A) 2 B) 3 C) 4 D) 5

184) The inner nuclear membrane contains specific .....that act as binding site?

A) Ribosome B) Protein C) mRNA D) tRNA

185) Bidirectional traffic between the cytosol and ....?

A) Inner membrane B) Outer membrane C) Protein D) All of these

186) Each complex has molecular mass of about .....?

A) 125 hundred B) 125 thousand C) 125 millions D) 125 billions

187) Each complex is composed of about.....different protein?

A) 20 B) 30 C) 40 D) 50

188) 17,000 daltons takes ....minutes to protein larger?

A) 2 B) 3 C) 4 D) 5

189) Nuclear pore contains a pathway for few diffusion .....nm in diameter and .....long?

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A) 9 , 14 B) 9 , 15 C) 9 , 16 D) 9 , 17

190) The nuclear envelop of a typically mammalian cell contains.....more complex?

A) 1000 – 2000 B) 2000 – 3000 C) 3000 – 4000 D) 4000 – 5000

191) The signal defined in numerous proteins by using .....technology?

A) DNA recombinant B) RNA recombinant C) both D) None

192) Immunoflourescence microphages showing the cellular location of ...virus ?

A) SV_{20} B) SV_{30} C) SV_{40} D) SV_{50}

193) .....repeat line the path through nuclear pore complex taken by the import receptor?

A) GF B) FG C) Cytosol D) a and b

194) The energy is provided by the hydrolysis of .....?

A) ATP B) ADP C) GDP D) GTP

195) How many subcompartment in mitochondria ?

A))One B) Two C) Three D) Four

196) Matrix signal sequences have propensity to fold into an amphipathic .....helix?

A) α B) β C) γ D) All of these

197)Chaperon protein belonging to the.....family?

A) hsp60 B) hsp70 C) hsp80 D) hsp90

198) TIM22 complex mediates .....of inner membrane protein?

A) Duplication B) Deletion C) Insertion D) Single gene

199) The N – terminal signal sequence of precursor protein is recognized by.....complex?

A) TIM B) TOM C) TIM22 D) TOM22

200) Protein transport require .....signal sequences?

A) One B) Two C) Three D) Four

201) The regions of ER that lack bound ribosome is called ?

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A) Rough ER B) ER membrane C) Smooth ER D) a an c

202) ER protein import is a co .....process?

A) Translational B) Transcriptional C) Transformational D) a and b

203) The SRP consist of ....different polypeptide chains ?

A) Three B) Four C) Five D) Six

204) The translocator .....complex forms a water filled pore in the membrane ?

A) Sec61 B) Sec62 C) Sec63 D) Sec64

205) The stop transfer sequence is a ....α helix membrane?

A) Single B) Double C) Triple D) b and c

Mcqs lec 159 – 168

206) Golgi apparatus are packed into small ....coated transport vesicles ?

A) COP11 B) CPO11 C) C11PO D) C11OP

207) Which protein are actively recrurited into....?

A) Repressor B) Cargo C) Receptor D) Resident

208) Which protein serve as receptor .....for packing some secondary protein into COP11 coated vesicles?

A) ERG1C51 B) ERG1C52 C) ERG1C53 D) ERG1C54

209) The ERG1C53 protein is a lectin that bind on factor .....?

A) 111 and 1V B) v and v11 C) V and v1 D) V and V111

210) Homotypic fusion and heterotypic fusion requires a set of matching .....?

A) SNAREs B) SANREs C) SANERs D) SNAERs

211) Resident ER membrane protein consist of .....lysines?

A) One B) Two C) Three D) Four

212) Signal sequences is called ?

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A) KKXX B) Bip C) KDEL D) a and c

213) Lysosomes contain .....types of hydrolytic enzyme?

A) 10 B) 20 C) 30 D) 40

214) Lysosome contain PH....?

A) 5.0 B) 6.0 C) 7.0 D) 8.0

215) Lysosome hydrolysis carry a unique marker in the form of .....groups?

A) M5P B) M6P C) M7P D) M8P

216) M6P receptor binds its specific oligosaccharide at PH....?

A) 4.5 B) 5.5 C) 6.5 D) 7.5

217) Phagocytosis involves the ingestion of .....?

A) Fluid B) Solutes C) a and b D) Microorganism

218) How many classes of WBs in mammals ?

A) Three B) Four C) Five D) Six

219) Phagcytosis partical contain a core of about....cholesterol molecule ?

A) 1400 B) 1500 C) 1600 D)1700

220) Skin regenerate every .....?

A) 10 years B) One week C) Two week D) 20 years

221) All organs have limited ability to regenerate ...?

A) 4 cm B) 3 cm C) 2 cm D) 1 cm

222) Every .....a person dies from a disease that can be treated with tissue replacement?

A) 30 sec B) 30 min C) 20 sec D) 20 min

223) Human blood discover.....?

A) Alexander Bogdanov Moscow B) Austrian Karl Landsteiner C) Joseph Murray D) Eduard Zirm

224) Embryonic stem cell are derived from the embryos .....?

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A) AA option B) BB option C) CC option D) DD option

225) Dusseldorf group performed clinical stem cell application for the first time...?

A) 2004 B) 2003 C) 2002 D) 2001

226) Which cells do not cause rejection?

a) WBCs b) RBCs c) adult stem cells d) customized stem cells

227) There is a growing evidence that bone marrow is derived stem cells which improve!

a) cardiac function b) muscle function c) Stumich function d) Both a and c

228) The evidence suggested by ____about bone marrow ;

a) Leri b) Robert c) irtaza

229) Stem cells right from ;

a) oxygenated blood b) deoxygenated blood c) cord blood d) both a and b

230) Cord blood have cells with features of____ stem cells;

A) adult embryonic C) customized D) both a and b

231) Fertilized egg result in the formation of :

A) somatic cells sex cells C) mature cells D) embryo

232) When cells taken from the embryos in embryonic stem cells as a result embryo

A) copied dead C) shrink. D) both a and b in some conditions

233) Macrophages scevange senescent RBCs regenerate about ;

A) 10 million 100 million C) 10 billion. D) 100 billion

Lec 169 – 177

234) In .....Bogdanov blood transfusion experiment?

A) 1924 B) 1925 C) 1926 D) 1927

235) Lenin´s sister..... was take part in Bogdanvo experiment?

A) Sadia B) Maria C) Nadia D) Shazia

236) In .....2014 the media reported on a blood transfusion that contain a protein called....?

A) March , GFD11 B) April , G11DF C) May , GDF11 D) June , G11FD

237) Which things should be used all time when working with chemicals or biological agent ?

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A) Lab coat B) Safety glasses C) Special gloves D) All of these

238) Which gas maintain the ph in incubator ?

A) Co_2 B) N C) O D) All of these

239) Microscope is used for.....?

A) RNA copies B) DNA copies C) a and b D) SRNA

240) In centrifuge mechine temperature is used ...?

A) 1000 – 15000◦C B) 1000 – 16000◦C C) 1000 – 17000◦C D) 1000 – 18000◦C

241)) In centrifuge mechine..... temperature is used for DNA extract?

A) 4◦C B) 18◦C C) 19◦C D) 20◦C

242) In water bath is used fix temperature .....?

A) 37◦C B) 32◦C C) 40◦C D) 4◦C

243) In incubators Co_2 is used ....?

A) 5 – 8 % B) 5 – 10 % C) 5 – 12 % D) 5 – 14 %

244) Ethanol .....is used in incubator?

A) 50 % B) 60 % C) 70 % D) 80 %

245) How many pore size of filter ?

A) 5 B) 4 C) 3 D 2

246) Two pore size are .....of filter?

A) 0.2um and 0.45um B) 0.2um and 0.46um C) 0.2um and 0.47um D) 0.2um and 0.48um

247) A vertebrate can have more than .....different cell types?

A) 80 B) 100 C) 120 D) 200

248) How many types of cell culture?

A) 4 B) 3 C) 2 D) 1

249) Cell lines have at least ....passage?

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A) One B) Two C) Three D) Four

250) Which type of cell culture is dissociated enzymatically?

A) Suspension B) Adherent C) a and b D) None

251) suspension cultures have density .....?

A) 2.0 to 2.5 B) 0.7 to 1.0 C) a and b D) None

252) In which phase DNA replicate?

A) M B) G1 C)G2 D) S

253) Which phase is maximum condensed ?

A) Prophase B) Metaphase C) Anaphase D) Telophase

254) Separation of chromatids is start in.....?

A) Prophase B) Metaphase C) Anaphase D) Telophase

255) In animal cell , invisible thread tightening between the ...poles?

A) Five B) Four C) Three D) Two

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